Leveraging GxE Interaction to Optimize Multi-Trait Selection in Lima Bean

AUTHOR
João Paulo Silva Pavan

PUBLISHED
June 11, 2026

Packages

```
library(reshape2)
library(ggplot2)
library(openxlsx)
library(asreml)
```

```
Online License checked out Thu Jun 11 13:01:52 2026
```

```
library(dplyr)
library(tibble)
library(metan)
library(writexl)
```

```
knitr::opts_chunk$set(echo = TRUE)
  knitr::opts_knit$set(root.dir = "../")
```

Lima Bean Data

- 3 Locations (Brazil) - Piracicaba - SP, Teresina - PI, Tianguá - CE,
- 40 Lima Bean Lines
- Randomized Complete Block Design (RCBD) - 3 Repetitions
- Plot usable area: 5m²
- Traits:

Trait	Acronym	Unit
Grain Yield	GY	kg/ha
Number days to flowering	NDF	days
Number days to maturity	NDM	days
One Hundred Seed Weight	OHSW	g
Plant Height	PH	cm
Pod Length	PL	mm

Trait	Acronym	Unit
Pod Number	PN	mm
Pod Thickness	PT	mm
Pod Width	PW	mm
Seed Length	SL	mm
Seed Thickness	ST	mm
Seed Width	SW	mm

Load the data

```
LB = read.xlsx("Data//LimaBean.xlsx")
LB$Block    <- as.factor(LB$Block)
LB$Env    <- as.factor(LB$Env)
LB$Genotype <- as.factor(LB$Genotype)
LB$GEN<- as.factor(LB$GEN)
str(LB)
```

```
'data.frame':	288 obs. of  16 variables:
 $ Env     : Factor w/ 3 levels "Piracicaba","Teresina",..: 3 2 2 2 2 2 3 2 2 2 ...
 $ Block   : Factor w/ 3 levels "1","2","3": 2 1 1 1 2 1 2 1 1 3 ...
 $ Genotype: Factor w/ 40 levels "H25_53","H25_54",..: 37 39 16 11 28 38 25 28 40 38 ...
 $ GEN     : Factor w/ 40 levels "L01","L02","L03",..: 37 39 16 11 28 38 25 28 40 38 ...
 $ NDF     : num  30 36 63 42 91 39 30 64 52 48 ...
 $ NDM     : num  71 125 125 125 125 125 91 133 125 93 ...
 $ PH      : num  41.4 33.4 33.2 29.4 38.5 ...
 $ PL      : num  NA 55.3 55 51.5 48.6 ...
 $ PW      : num  NA 13.9 12.9 12.4 13.5 ...
 $ PT      : num  NA 7.96 8.81 6.72 7.19 8.69 9.15 7.42 6.12 8.28 ...
 $ SL      : num  NA 10.78 10.17 9.44 10.64 ...
 $ SW      : num  NA 6.82 6.62 6.2 7.37 7.8 9.01 7.28 7.6 7.48 ...
 $ ST      : num  NA 4.64 4.55 4.75 5.35 5.5 5.2 5.15 4.92 5.46 ...
 $ PN      : num  NA 22 5 37 35 32 25 48 62 62 ...
 $ GY      : num  NA 15.3 21.4 23.1 24.5 ...
 $ OHSW    : num  NA 24.9 10.7 26.5 28.8 35.5 46 25.2 27.1 35.5 ...
```

Histograms

```
hist(LB$NDF)
hist(LB$NDM)
hist(LB$PH)
hist(LB$PL)
hist(LB$PW)
hist(LB$PT)
hist(LB$SL)
hist(LB$SW)
hist(LB$ST)
```

```
hist(LB$PN)
hist(LB$GY)
hist(LB$OHSW)
```

Anova

```
aov = anova_joint(LB,env = Env, gen = GEN, rep = Block,resp = everything())
```

Single environment analyses

```
data.list = split(LB, f = LB$Env)
vccomp=list()
herit=list()
j="Piracicaba"
for (j in names(data.list)) {

   x = droplevels(data.list[[j]])
  cat("====> Environment:", j, fill = TRUE)

st <- gamem(x,
             gen = GEN,
             rep =  Block,
             resp = everything())
st
vccomp[[j]]=get_model_data(st, "vcomp")
vccomp[[j]]$env=j
herit[[j]]=get_model_data(st, "h2")
herit[[j]]$env=j
}
```

```
====> Environment: Piracicaba
Evaluating trait NDF |====                                          | 8% 00:00:00
Evaluating trait NDM |=======                                       | 17% 00:00:00
Evaluating trait PH |===========                                    | 25% 00:00:00
Evaluating trait PL |===============                                | 33% 00:00:00
Evaluating trait PW |==================                             | 42% 00:00:00
Evaluating trait PT |======================                         | 50% 00:00:00
Evaluating trait SL |==========================                     | 58% 00:00:00
Evaluating trait SW |=============================                  | 67% 00:00:00
Evaluating trait ST |=================================              | 75% 00:00:00
Evaluating trait PN |=====================================          | 83% 00:00:01
Evaluating trait GY |========================================       | 92% 00:00:01
Evaluating trait OHSW |==========================================| 100% 00:00:01

---------------------------------------------------------------------------
P-values for Likelihood Ratio Test of the analyzed traits
---------------------------------------------------------------------------
    model    NDF   NDM   PH      PL       PW     PT      SL      SW   ST     PN
 Complete     NA    NA   NA      NA       NA     NA      NA      NA   NA     NA
 Genotype 0.0413 0.683 0.03 0.00124 2.05e-06 0.0766 9.6e-06 9.2e-07 0.77 0.0358
```

```
    GY     OHSW
    NA       NA
 0.113 1.12e-10
---------------------------------------------------------------------------
Variables with nonsignificant Genotype effect
NDM PT ST GY
---------------------------------------------------------------------------

====> Environment: Teresina
Evaluating trait NDF |====                                           | 8% 00:00:00
Evaluating trait NDM |=======                                        | 17% 00:00:00
Evaluating trait PH |===========                                     | 25% 00:00:00
Evaluating trait PL |===============                                 | 33% 00:00:00
Evaluating trait PW |==================                              | 42% 00:00:00
Evaluating trait PT |======================                          | 50% 00:00:00
Evaluating trait SL |==========================                      | 58% 00:00:00
Evaluating trait SW |=============================                   | 67% 00:00:00
Evaluating trait ST |=================================               | 75% 00:00:00
Evaluating trait PN |=====================================           | 83% 00:00:00
Evaluating trait GY |========================================        | 92% 00:00:01
Evaluating trait OHSW |==========================================| 100% 00:00:01

---------------------------------------------------------------------------
P-values for Likelihood Ratio Test of the analyzed traits
---------------------------------------------------------------------------
    model      NDF     NDM       PH      PL       PW   PT       SL      SW
 Complete       NA      NA       NA      NA       NA   NA       NA      NA
 Genotype 8.89e-11 0.00612 1.86e-06 0.0923 2.73e-05 0.36 0.000835 8.9e-06
      ST       PN       GY     OHSW
      NA       NA       NA       NA
 0.00292 3.22e-05 2.82e-05 0.000647
---------------------------------------------------------------------------
Variables with nonsignificant Genotype effect
PL PT
---------------------------------------------------------------------------

====> Environment: Tiangua
Evaluating trait NDF |====                                           | 8% 00:00:00
Evaluating trait NDM |=======                                        | 17% 00:00:00
Evaluating trait PH |===========                                     | 25% 00:00:00

Evaluating trait PL |===============                                 | 33% 00:00:00

Evaluating trait PW |==================                              | 42% 00:00:00

Evaluating trait PT |======================                          | 50% 00:00:00

Evaluating trait SL |==========================                      | 58% 00:00:00

Evaluating trait SW |=============================                   | 67% 00:00:00

Evaluating trait ST |=================================               | 75% 00:00:00

Evaluating trait PN |=====================================           | 83% 00:00:00
```

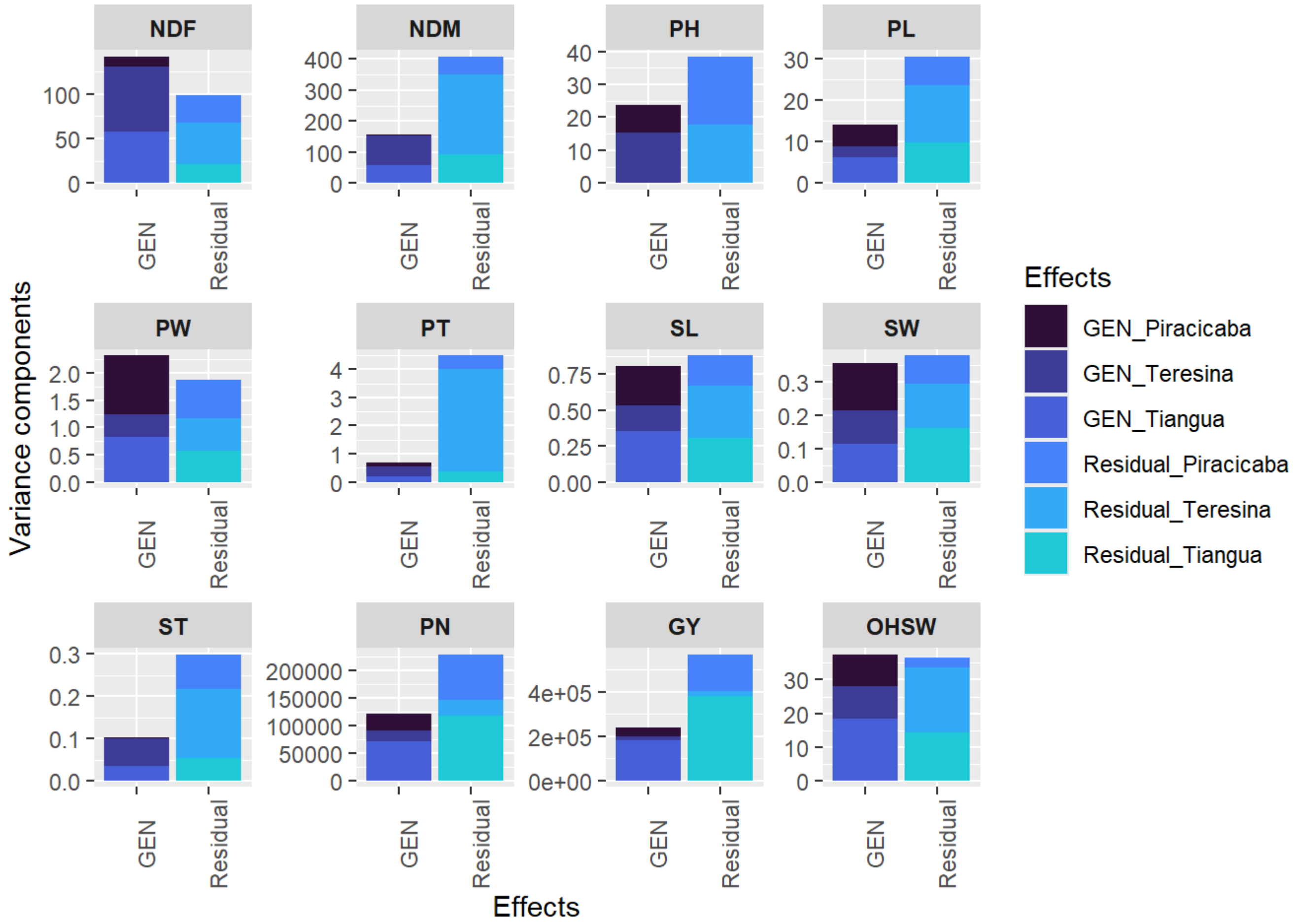
```
Evaluating trait GY |========================================    | 92% 00:00:00

Evaluating trait OHSW |==========================================| 100% 00:00:01

---------------------------------------------------------------------------
P-values for Likelihood Ratio Test of the analyzed traits
---------------------------------------------------------------------------
    model      NDF      NDM       PH       PL       PW       PT       SL
 Complete       NA       NA       NA       NA       NA       NA       NA
 Genotype 4.59e-14 0.000249 3.18e-44 0.000304 8.99e-09 0.000596 2.38e-07
       SW      ST     PN      GY     OHSW
       NA      NA     NA      NA       NA
 0.000101 0.00037 4e-04 0.00262 5.17e-08
---------------------------------------------------------------------------
All variables with significant (p < 0.05) genotype effect
```

Variance components

```
custom_palette=viridis::turbo(n = 19)
vc=do.call(rbind,vccomp)
colnames(vc)
```

```
 [1] "Group" "NDF"   "NDM"   "PH"    "PL"    "PW"    "PT"    "SL"    "SW"   
[10] "ST"    "PN"    "GY"    "OHSW"  "env"  
```

```
traits=colnames(vc[,2:13])
vc_long <- reshape2::melt(vc, measure.vars = traits, variable.name = "trait")

vc_long$effect =paste(vc_long$Group,vc_long$env, sep = "_")

vcp <- vc_long |>
  ggplot(aes(x = Group, y = value, fill = effect)) +
  geom_col(position = "stack", just = 0.5) +
  theme(axis.text.x = element_text(angle = 90),
        strip.text = element_text(face = "bold")) +
  facet_wrap(~trait, scales = "free") +
  labs(x = "Effects", y = "Variance components", fill = "Effects") +
  scale_fill_manual(
    values = custom_palette,
  )
vcp
```



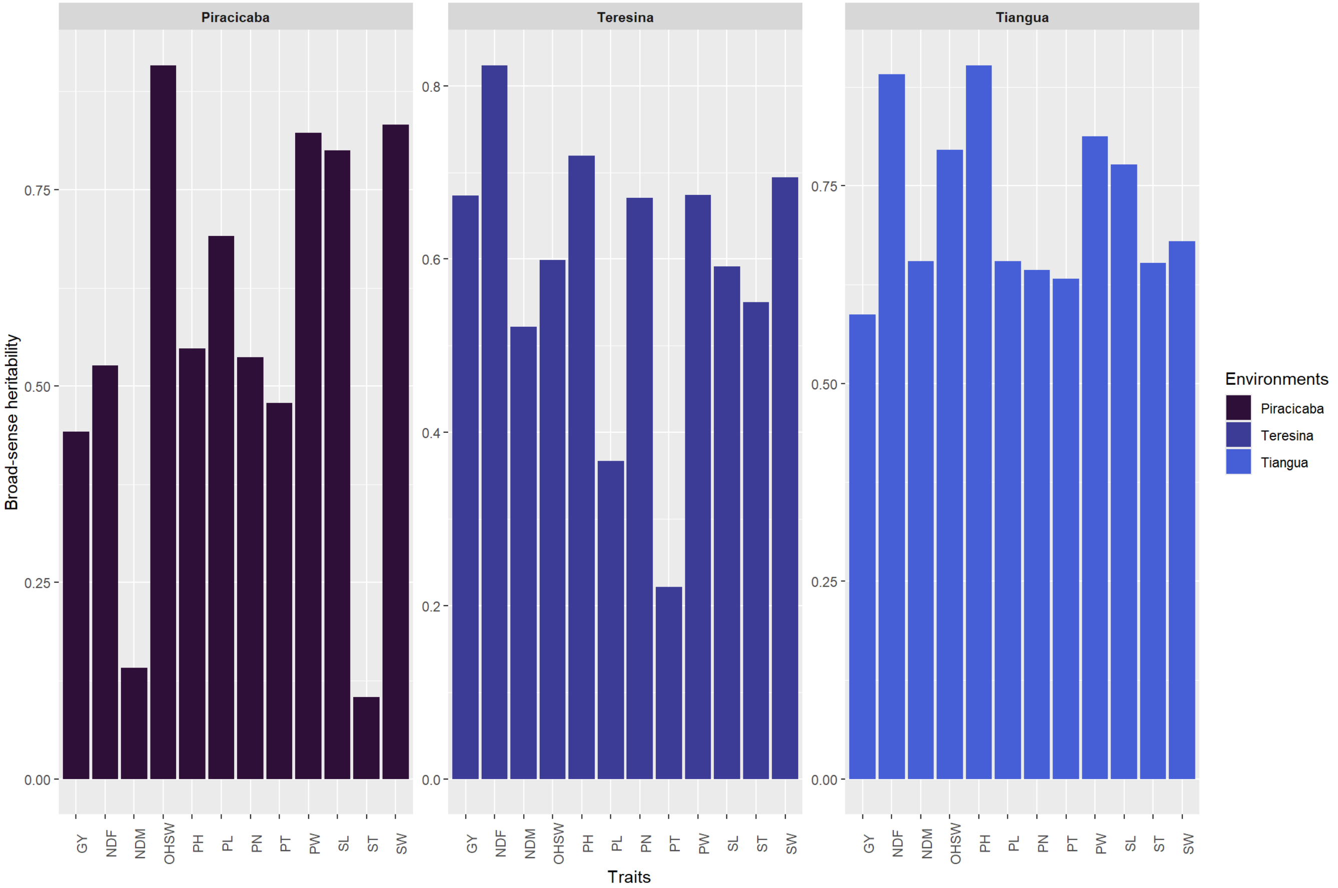

```
ggsave(plot=vcp,device = "pdf",filename ="Plots/varcomp.pdf",dpi = "retina",height = 6,width =
ggsave(plot=vcp,device = "png",filename ="Plots/varcomp.png",dpi = "retina",height = 6,width =
```

Broad-sense Heritability

```
custom_palette=viridis::turbo(n = 19)
her=do.call(rbind,herit)
colnames(her)
```

```
[1] "VAR" "h2"  "env"
```

```
heritp <- her |>
  ggplot(aes(x = VAR, y = h2, fill = env)) +
  geom_col(position = "stack", just = 0.5) +
  theme(axis.text.x = element_text(angle = 90),
        strip.text = element_text(face = "bold")) +
  facet_wrap(~env, scales = "free") +
  labs(x = "Traits", y = "Broad-sense heritability", fill = "Environments") +
  scale_fill_manual(
    values = custom_palette)
heritp
```




```
ggsave(plot=heritp,device = "pdf",filename ="Plots/herit.pdf",dpi = "retina",height = 6,width
ggsave(plot=heritp,device = "png",filename ="Plots/herit.png",dpi = "retina",height = 6,width
```

Mixed-effect models

```
mixedmodel <- gamem_met(LB,env = Env, gen = GEN, rep = Block,random = "gen" ,resp = everything
```

```
Evaluating trait NDF |====                                      | 8% 00:00:01
Evaluating trait NDM |=======                                   | 17% 00:00:02
Evaluating trait PH |===========                                | 25% 00:00:02

Evaluating trait PL |===============                            | 33% 00:00:03

Evaluating trait PW |==================                         | 42% 00:00:04

Evaluating trait PT |======================                     | 50% 00:00:04

Evaluating trait SL |==========================                 | 58% 00:00:05

Evaluating trait SW |=============================              | 67% 00:00:05

Evaluating trait ST |=================================          | 75% 00:00:06

Evaluating trait PN |=====================================      | 83% 00:00:06
```

```
Evaluating trait GY |=======================================     | 92% 00:00:07

Evaluating trait OHSW |==========================================| 100% 00:00:07

---------------------------------------------------------------------------
P-values for Likelihood Ratio Test of the analyzed traits
---------------------------------------------------------------------------
    model      NDF    NDM       PH      PL       PW      PT       SL       SW
 COMPLETE       NA     NA       NA      NA       NA      NA       NA       NA
      GEN 3.43e-06 0.0788 4.52e-01 0.34753 3.70e-05 0.00102 8.53e-06 0.000199
  GEN:ENV 8.16e-06 0.0120 2.74e-07 0.00107 2.86e-05 1.00000 1.57e-02 0.000918
      ST       PN      GY     OHSW
      NA       NA      NA       NA
 0.00083 2.65e-01 0.21557 3.40e-07
 0.46110 5.04e-05 0.00108 3.51e-02
---------------------------------------------------------------------------
Variables with nonsignificant GxE interaction
PT ST
---------------------------------------------------------------------------
```

```r
library(dplyr)
library(tidyr)
library(purrr)

ne <- length(unique(LB$Env))
nr <- length(unique(LB$Block))

varcomp <- map_dfr(names(mixedmodel), function(tr){

  if(!"random" %in% names(mixedmodel[[tr]]))
    return(NULL)
  vc <- mixedmodel[[tr]]$random
  tibble(
    Trait = tr,
    sigma_g  = vc$Variance[vc$Group == "GEN"],
    sigma_ge = vc$Variance[vc$Group == "GEN:ENV"],
    sigma_e  = vc$Variance[vc$Group == "Residual"]
  )
})
genpar <- get_model_data(mixedmodel) %>%
  pivot_longer(
    cols = -Parameters,
    names_to = "Trait",
    values_to = "Value") %>%
  pivot_wider(
    names_from = Parameters,
    values_from = Value)

herit_table <- varcomp %>%
  left_join(genpar, by = "Trait") %>%
  dplyr::mutate(

    H2_individual =
      sigma_g /
```

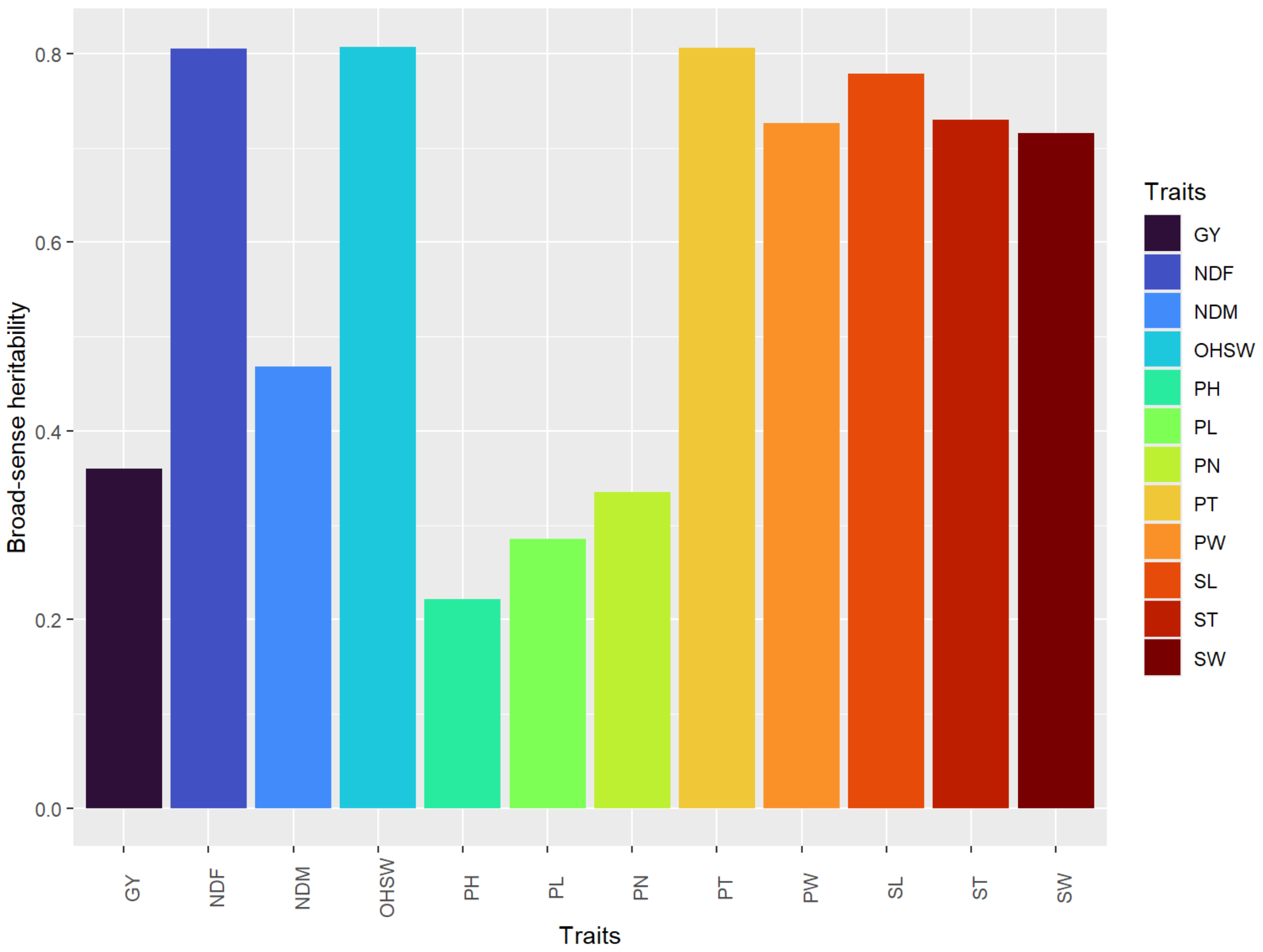
```
      (sigma_g + sigma_ge + sigma_e),

    H2_MET =
      sigma_g /
      (sigma_g +
         sigma_ge/ne +
         sigma_e/(ne*nr)),

    H2mg_metan = h2mg) %>%
  dplyr::select(
    Trait,
    sigma_g,
    sigma_ge,
    sigma_e,
    H2_individual,
    H2_MET,
    H2mg_metan,
    Accuracy,
    GEIr2,
    rge,
    CVg,
    CVr,
    `CV ratio`)
herit_table
```

```
# A tibble: 12 × 13
   Trait    sigma_g    sigma_ge sigma_e H2_individual H2_MET H2mg_metan Accuracy
   <chr>      <dbl>       <dbl>   <dbl>         <dbl>  <dbl>      <dbl>    <dbl>
 1 NDF      38.9       16.9     3.41e+1        0.432  0.805      0.805    0.897
 2 NDM      25.5       35.7     1.54e+2        0.119  0.468      0.468    0.684
 3 PH        1.07       7.21    1.21e+1        0.0524 0.222      0.222    0.471
 4 PL        0.947      3.51    1.08e+1        0.0620 0.286      0.286    0.534
 5 PW        0.423      0.273   6.15e-1        0.322  0.726      0.726    0.852
 6 PT        0.728      0       1.57e+0        0.316  0.806      0.806    0.898
 7 SL        0.199      0.0676  3.09e-1        0.346  0.778      0.778    0.882
 8 SW        0.0727     0.0430  1.32e-1        0.294  0.715      0.715    0.846
 9 ST        0.0376     0.00639 1.06e-1        0.250  0.729      0.729    0.854
10 PN     9271.     31235.      7.21e+4        0.0824 0.335      0.335    0.579
11 GY    22388.     58803.      1.81e+5        0.0853 0.360      0.360    0.600
12 OHSW     10.0        2.56    1.38e+1        0.379  0.807      0.807    0.898
# ℹ 5 more variables: GEIr2 <dbl>, rge <dbl>, CVg <dbl>, CVr <dbl>,
#   `CV ratio` <dbl>
```

Broad-sense Heritability MET

```
herit_met=get_model_data(mixedmodel, "h2")

custom_palette=viridis::turbo(n = 12)
heritp2 <- herit_met |>
  ggplot(aes(x = VAR, y = h2, fill=VAR)) +
  geom_col(position = "stack", just = 0.5) +
  theme(axis.text.x = element_text(angle = 90),
```

```
        strip.text = element_text(face = "bold")) +
  labs(x = "Traits", y = "Broad-sense heritability",fill = "Traits") +
  scale_fill_manual(
    values = custom_palette,
  )
heritp2
```




```
ggsave(plot=heritp2,device = "pdf",filename ="Plots/herit_met.pdf",dpi = "retina",height = 6,w
ggsave(plot=heritp2,device = "png",filename ="Plots/herit_met.png",dpi = "retina",height = 6,w
```

Culllis Heritability

```
traits <- c("PH", "NDF", "NDM", "OHSW", "PN", "GY")
library(sommer)
library(purrr)
library(dplyr)
calc_cullis <- function(tr){
  mod <- mmer(
    fixed  = as.formula(
      paste0(tr," ~ Env + Block:Env")
    ),
    random = ~ GEN + GEN:Env,
```

```
    rcov   = ~ units,
    data   = LB
  )
  vg <- mod$sigma$GEN
  C22 <- mod$PevU$GEN[[tr]]
  n_g <- nrow(C22)
  trC22 <- sum(diag(C22))

  av2 <- 2/n_g *
    (trC22 -
      (sum(C22)-trC22)/(n_g-1))

  H2 <- 1 - av2/(2*vg)

  tibble(
    Trait = tr,
    Cullis_H2 = as.numeric(H2))

}
Cullis_all <- map_dfr(
  traits,
  calc_cullis)
```

```
iteration    LogLik     wall    cpu(sec)   restrained
    1      -9.79748   13:2:8      0          0
    2      -6.85572   13:2:8      0          0
    3      -6.17192   13:2:8      0          0
    4      -6.11791   13:2:8      0          0
    5      -6.11748   13:2:8      0          0

iteration    LogLik     wall    cpu(sec)   restrained
    1      -38.7664   13:2:8      0          0
    2      -26.3316   13:2:8      0          0
    3      -23.5492   13:2:8      0          0
    4      -23.3172   13:2:9      1          0
    5      -23.3144   13:2:9      1          0
    6      -23.3144   13:2:9      1          0

iteration    LogLik     wall    cpu(sec)   restrained
    1      -80.8335   13:2:9      0          0
    2      -80.7879   13:2:9      0          0
    3      -80.7715   13:2:9      0          0
    4      -80.7691   13:2:9      0          0
    5      -80.769   13:2:9      0          0

iteration    LogLik     wall    cpu(sec)   restrained
    1      -70.2583   13:2:9      0          0
    2      -64.6976   13:2:9      0          0
    3      -63.7078   13:2:9      0          0
    4      -63.6435   13:2:9      0          0
    5      -63.6432   13:2:9      0          0

iteration    LogLik     wall    cpu(sec)   restrained
    1      -76.0017   13:2:10      0          0
```

```
    2      -74.8831   13:2:10      0           0
    3      -74.5778   13:2:10      0           0
    4      -74.5462   13:2:10      0           0
    5      -74.5456   13:2:10      0           0

iteration    LogLik     wall    cpu(sec)   restrained
    1      -47.403   13:2:10      0           0
    2      -46.9832   13:2:10      0           0
    3      -46.8571   13:2:10      0           0
    4      -46.8422   13:2:10      0           0
    5      -46.8419   13:2:10      0           0
```

```
Cullis_all
```

```
# A tibble: 6 × 2
  Trait Cullis_H2
  <chr>     <dbl>
1 PH        0.182
2 NDF       0.752
3 NDM       0.403
4 OHSW      0.754
5 PN        0.281
6 GY        0.303
```

LRT

```
GYLRT = mixedmodel$GY$LRT
OHSWLRT = mixedmodel$OHSW$LRT
NDFLRT = mixedmodel$NDF$LRT
NDMLRT = mixedmodel$NDM$LRT
PHLRT = mixedmodel$PH$LRT
PNLRT = mixedmodel$PN$LRT
LRT = list(
  GY_LRT    = GYLRT,
  OHSW_LRT = OHSWLRT,
  NDF_LRT   = NDFLRT,
  NDM_LRT   = NDMLRT,
  PH_LRT    = PHLRT,
  PN_LRT    = PNLRT)
LRT
```

```
$GY_LRT
              model npar  logLik    AIC     LRT Df Pr(>Chisq)
<none>            1   12 -2131.5 4286.9
(1 | GEN)         2   11 -2132.2 4286.5  1.5336  1   0.215571
(1 | GEN:ENV)     3   11 -2136.8 4295.6 10.6932  1   0.001075 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$OHSW_LRT
              model npar  logLik    AIC    LRT Df Pr(>Chisq)
<none>            1   12 -823.50 1671.0
(1 | GEN)         2   11 -836.50 1695.0 26.008  1    3.4e-07 ***
```

```
(1 | GEN:ENV)     3   11 -825.72 1673.4  4.441  1    0.03509 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$NDF_LRT
              model npar  logLik    AIC    LRT Df Pr(>Chisq)
<none>            1   12 -974.03 1972.1
(1 | GEN)         2   11 -984.81 1991.6 21.559  1  3.432e-06 ***
(1 | GEN:ENV)     3   11 -983.98 1990.0 19.900  1  8.159e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$NDM_LRT
              model npar  logLik    AIC    LRT Df Pr(>Chisq)
<none>            1   12 -1148.4 2320.9
(1 | GEN)         2   11 -1150.0 2322.0 3.0888  1    0.07883 .
(1 | GEN:ENV)     3   11 -1151.6 2325.2 6.3046  1    0.01204 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$PH_LRT
              model npar  logLik    AIC     LRT Df Pr(>Chisq)
<none>            1   12 -811.05 1646.1
(1 | GEN)         2   11 -811.33 1644.7  0.5646  1     0.4524
(1 | GEN:ENV)     3   11 -824.26 1670.5 26.4263  1  2.738e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$PN_LRT
              model npar  logLik    AIC     LRT Df Pr(>Chisq)
<none>            1   12 -2009.8 4043.6
(1 | GEN)         2   11 -2010.4 4042.8  1.2403  1     0.2654
(1 | GEN:ENV)     3   11 -2018.0 4058.0 16.4317  1  5.044e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
write_xlsx(LRT, "LRT_LB.xlsx")
```

BLUPs

```
BLUPgen = list(
  GY_BLUP   = mixedmodel$GY$BLUPgen,
  OHSW_BLUP = mixedmodel$OHSW$BLUPgen,
  NDF_BLUP  = mixedmodel$NDF$BLUPgen,
  NDM_BLUP  = mixedmodel$NDM$BLUPgen,
  PH_BLUP   = mixedmodel$PH$BLUPgen,
  PN_BLUP   = mixedmodel$PN$BLUPgen)
head(BLUPgen)
```

```
$GY_BLUP
# A tibble: 40 × 7
    Rank GEN       Y BLUPg Predicted    LL    UL
   <dbl> <fct> <dbl> <dbl>     <dbl> <dbl> <dbl>
```

```
 1     1 L06   1380. 164.       913.  612. 1214.
 2     2 L32   1304. 144.       892.  591. 1193.
 3     3 L18   1113. 122.       871.  570. 1172.
 4     4 L24   1094. 115.       864.  563. 1165.
 5     5 L30   1085. 112.       861.  560. 1162.
 6     6 L07   1043.  96.8      846.  544. 1147.
 7     7 L17    893.  96.7      845.  544. 1146.
 8     8 L33    976.  72.7      821.  520. 1122.
 9     9 L05   1020.  66.2      815.  514. 1116.
10    10 L23    923.  53.5      802.  501. 1103.
# ℹ 30 more rows

$OHSW_BLUP
# A tibble: 40 × 7
    Rank GEN       Y BLUPg Predicted    LL    UL
   <dbl> <fct> <dbl> <dbl>     <dbl> <dbl> <dbl>
 1     1 L40    38.6  5.34      37.5  34.3  40.7
 2     2 L14    39.9  5.32      37.5  34.3  40.7
 3     3 L25    37.2  4.18      36.4  33.2  39.6
 4     4 L21    36.5  3.63      35.8  32.6  39.0
 5     5 L26    36.1  3.30      35.5  32.3  38.7
 6     6 L27    35.1  2.48      34.7  31.5  37.9
 7     7 L23    34.9  2.38      34.6  31.4  37.8
 8     8 L30    34.8  2.27      34.4  31.2  37.7
 9     9 L18    34.3  1.87      34.0  30.8  37.3
10    10 L20    34.2  1.79      34.0  30.8  37.2
# ℹ 30 more rows

$NDF_BLUP
# A tibble: 40 × 7
    Rank GEN       Y BLUPg Predicted    LL    UL
   <dbl> <fct> <dbl> <dbl>     <dbl> <dbl> <dbl>
 1     1 L23    55.3  9.73      53.3  47.0  59.7
 2     2 L28    55.3  9.73      53.3  47.0  59.7
 3     3 L16    51.6  6.69      50.3  43.9  56.7
 4     4 L26    51.2  6.42      50.0  43.7  56.4
 5     5 L30    51.1  6.33      49.9  43.6  56.3
 6     6 L19    50.8  6.06      49.7  43.3  56.0
 7     7 L22    50.4  5.79      49.4  43.0  55.8
 8     8 L27    50.3  5.70      49.3  43.0  55.7
 9     9 L18    50.1  5.52      49.1  42.8  55.5
10    10 L40    49.9  5.34      49.0  42.6  55.3
# ℹ 30 more rows

$NDM_BLUP
# A tibble: 40 × 7
    Rank GEN       Y BLUPg Predicted    LL    UL
   <dbl> <fct> <dbl> <dbl>     <dbl> <dbl> <dbl>
 1     1 L11   101    6.41      94.2  85.2 103.
 2     2 L40   101.   5.70      93.5  84.4 103.
 3     3 L23    98.9  4.87      92.6  83.6 102.
 4     4 L28    98.3  4.61      92.4  83.4 101.
 5     5 L19    97.6  4.25      92.0  83.0 101.
 6     6 L25    96.1  3.57      91.3  82.3 100.
```

```
 7      7 L16    95.9  3.47      91.2  82.2 100. 
 8      8 L18    95.6  3.31      91.1  82.1 100. 
 9      9 L22    94.8  2.95      90.7  81.7  99.8
10     10 L38    91.2  2.77      90.5  81.5  99.6
# ℹ 30 more rows

$PH_BLUP
# A tibble: 40 × 7
    Rank GEN       Y BLUPg Predicted    LL    UL
   <dbl> <fct> <dbl> <dbl>     <dbl> <dbl> <dbl>
 1     1 L28    48.7 1.10       43.6  41.2  46.0
 2     2 L22    46.5 0.628      43.2  40.8  45.5
 3     3 L27    46.4 0.603      43.1  40.7  45.5
 4     4 L23    46.3 0.573      43.1  40.7  45.5
 5     5 L30    46.2 0.542      43.1  40.7  45.5
 6     6 L19    46.1 0.519      43.0  40.7  45.4
 7     7 L26    45.7 0.443      43.0  40.6  45.4
 8     8 L17    47.5 0.420      42.9  40.6  45.3
 9     9 L40    45.5 0.395      42.9  40.5  45.3
10     10 L25   45.4 0.382      42.9  40.5  45.3
# ℹ 30 more rows

$PN_BLUP
# A tibble: 40 × 7
    Rank GEN       Y BLUPg Predicted    LL    UL
   <dbl> <fct> <dbl> <dbl>     <dbl> <dbl> <dbl>
 1     1 L32    963  104.       679.  480.  878.
 2     2 L05    847.  74.5      650.  451.  849.
 3     3 L04    844.  73.6      649.  450.  848.
 4     4 L02    832.  70.6      646.  447.  845.
 5     5 L07    812.  69.7      645.  446.  844.
 6     6 L33    803.  66.8      642.  443.  841.
 7     7 L24    775.  57.4      633.  434.  831.
 8     8 L17    744.  53.6      629.  430.  828.
 9     9 L30    739   45.3      620.  422.  819.
10     10 L18   734.  43.7      619.  420.  818.
# ℹ 30 more rows
```

```
write_xlsx(BLUPgen, "BLUPgen_LB.xlsx")
```

```
sommer_models <- setNames(
  map(traits, \(tr)
    mmer(
      fixed  = as.formula(
        paste0(tr, " ~ Env + Block:Env")
      ),
      random = ~ GEN + GEN:Env,
      rcov   = ~ units,
      data   = LB
    )
  ),
  traits)
```

```
iteration    LogLik     wall    cpu(sec)   restrained
    1      -9.79748   13:2:11      0          0
    2      -6.85572   13:2:11      0          0
    3      -6.17192   13:2:11      0          0
    4      -6.11791   13:2:11      0          0
    5      -6.11748   13:2:11      0          0

iteration    LogLik     wall    cpu(sec)   restrained
    1      -38.7664   13:2:12      1          0
    2      -26.3316   13:2:12      1          0
    3      -23.5492   13:2:12      1          0
    4      -23.3172   13:2:12      1          0
    5      -23.3144   13:2:12      1          0
    6      -23.3144   13:2:12      1          0

iteration    LogLik     wall    cpu(sec)   restrained
    1      -80.8335   13:2:12      0          0
    2      -80.7879   13:2:12      0          0
    3      -80.7715   13:2:12      0          0
    4      -80.7691   13:2:12      0          0
    5      -80.769   13:2:12      0          0

iteration    LogLik     wall    cpu(sec)   restrained
    1      -70.2583   13:2:12      0          0
    2      -64.6976   13:2:12      0          0
    3      -63.7078   13:2:13      1          0
    4      -63.6435   13:2:13      1          0
    5      -63.6432   13:2:13      1          0

iteration    LogLik     wall    cpu(sec)   restrained
    1      -76.0017   13:2:13      0          0
    2      -74.8831   13:2:13      0          0
    3      -74.5778   13:2:13      0          0
    4      -74.5462   13:2:13      0          0
    5      -74.5456   13:2:13      0          0

iteration    LogLik     wall    cpu(sec)   restrained
    1      -47.403   13:2:13      0          0
    2      -46.9832   13:2:13      0          0
    3      -46.8571   13:2:13      0          0
    4      -46.8422   13:2:13      0          0
    5      -46.8419   13:2:13      0          0
```

```r
sommer_blups <- map(
  traits,
  function(tr){

    data.frame(
      GEN = gsub("GEN", "", names(sommer_models[[tr]]$U$GEN[[tr]])),
      BLUP_sommer = as.numeric(
        sommer_models[[tr]]$U$GEN[[tr]]
      )
    )
```

```
  }
)
names(sommer_blups) <- traits
```

```
#Compare BLUPs- gamem_met and mmer
library(dplyr)

compare_blups <- map_dfr(
  traits,
  function(tr){

    gamem_blup <-
      mixedmodel[[tr]]$BLUPgen |>
      dplyr::select(GEN, BLUPg)

    sommer_blup <- sommer_blups[[tr]]

    comp <- left_join(
      gamem_blup,
      sommer_blup,
      by = "GEN")
    tibble(
      Trait = tr,
      Correlation =
        cor(
          comp$BLUPg,
          comp$BLUP_sommer
        ),
      Max_Difference =
        max(
          abs(
            comp$BLUPg -
            comp$BLUP_sommer
          )
        )
    )

  }
)
compare_blups
```

```
# A tibble: 6 × 3
  Trait Correlation Max_Difference
  <chr>       <dbl>          <dbl>
1 PH          1.000      0.0000319
2 NDF         1.000      0.000226
3 NDM         1.000      0.00106
4 OHSW        1.000      0.0000478
5 PN          1.000      0.101
6 GY          1.000      0.0933
```

```
library(dplyr)
library(tibble)
```

```
ebvMat <- data.frame(
  GEN = mixedmodel$GY$BLUPgen$GEN,
  GY   = mixedmodel$GY$BLUPgen$Predicted,
  OHSW = mixedmodel$OHSW$BLUPgen$Predicted,
  NDF  = mixedmodel$NDF$BLUPgen$Predicted,
  NDM  = mixedmodel$NDM$BLUPgen$Predicted,
  PH   = mixedmodel$PH$BLUPgen$Predicted,
  PN   = mixedmodel$PN$BLUPgen$Predicted
) |>
  column_to_rownames("GEN") |>
  as.matrix()
dim(ebvMat)
```

```
[1] 40  6
```

```
head(ebvMat)
```

```
          GY     OHSW      NDF      NDM       PH       PN
L06 913.1034 37.51991 53.34655 94.18319 43.62356 678.7407
L32 892.2353 37.49419 53.34655 93.47780 43.15325 649.6380
L18 870.6745 36.36019 50.30658 92.64571 43.12811 648.8006
L24 863.9895 35.80589 50.03834 92.38568 43.09755 645.7856
L30 860.5489 35.47762 49.94893 92.02164 43.06723 644.8933
L07 845.5058 34.66053 49.68070 91.34557 43.04431 641.9544
```

```
save(ebvMat,file = "Data/ebvMat.rda")
```

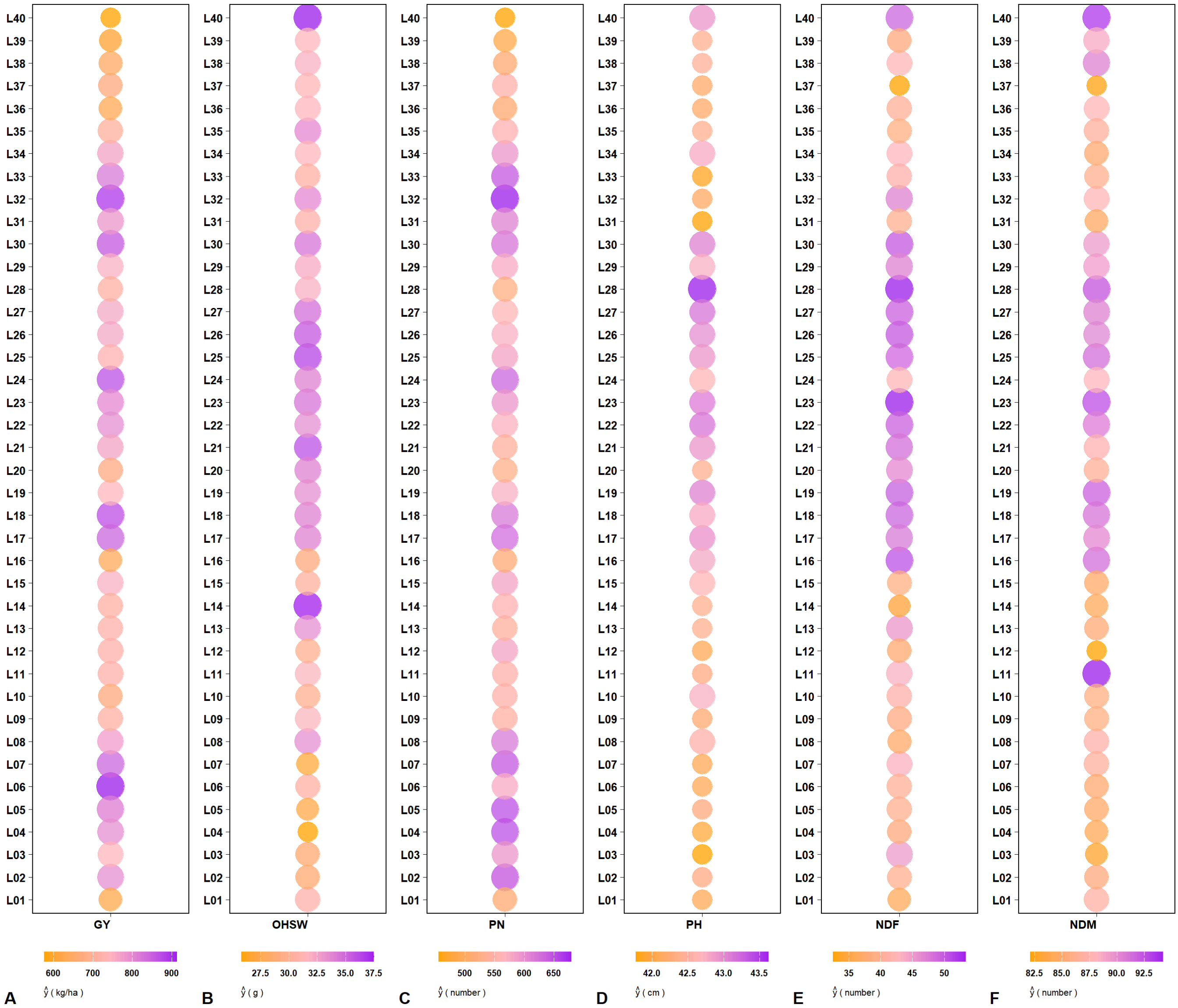
```
trait_units <- c(
  GY   = "kg/ha",
  OHSW = "g",
  PH   = "cm",
  PN   = "number",
  NDF  = "number",
  NDM  = "number")
plot_trait_heatmap <- function(x, trait_name) {
  gen_order <- sprintf("L%02d", 1:40)
  df <- x[[trait_name]]$BLUPgen %>%
    filter(GEN %in% gen_order) %>%
    dplyr::mutate(
      GEN = factor(GEN, levels = gen_order),
      Trait = trait_name
    )
  ggplot(df, aes(x = GEN, y = Trait, fill = Predicted)) +
    geom_tile() +
    scale_fill_gradientn(
      colours = c("orange", "#E37383", "purple"),
      name = bquote(hat(y)~"("~.(trait_units[[trait_name]])~")")) +
    labs(x = NULL, y = NULL) +
    theme_minimal(base_size = 14) +
    theme(
      axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1, face = "bold"),
```

```
      axis.text.y = element_text(angle = 90, face = "bold"),
      legend.position = "right",
      panel.grid = element_blank())
}
traits <- c("GY", "OHSW", "PH", "PN", "NDF", "NDM")
BLUP_heatmaps <- lapply(traits, function(trait) {
  plot_trait_heatmap(mixedmodel, trait)
})
library(ggpubr)
blups_heatmap_panel <- ggarrange(
  plotlist = BLUP_heatmaps,
  ncol = 2, nrow = 3,
  labels = c("A", "B", "C", "D", "E", "F"), font.label = list(size = 10, face = "bold"),
  align = "v")
ggsave(
  "BLUPs_LB.pdf",
  blups_heatmap_panel,
  width = 14,
  height = 12,
  dpi = 600)
```

```
trait_units <- c(
  GY   = "kg/ha",
  OHSW = "g",
  PH   = "cm",
  PN   = "number",
  NDF  = "number",
  NDM  = "number")
plot_trait_dot <- function(x, trait_name){
  gen_order <- sprintf("L%02d", 1:40)
  df <- x[[trait_name]]$BLUPgen %>%
    filter(GEN %in% gen_order) %>%
    dplyr::mutate(
      GEN = factor(GEN, levels = gen_order),
      Trait = trait_name)
  ggplot(df, aes(x = Trait, y = GEN)) +
    geom_point(
      aes(
        size  = round(abs(Predicted)),
        color = Predicted),
      alpha = 0.75) +
    scale_color_gradientn(
      colours = c("orange", "lightpink", "purple"),
      name = bquote(hat(y)~"("~.(trait_units[[trait_name]])~")"),
      guide = guide_colorbar(
    barheight = unit(0.3, "cm"),
    barwidth = unit(4, "cm"))) +

    scale_size(range = c(8,11), guide = "none") +

    labs(
      x = NULL,
      y = NULL) +
```

```
    theme_linedraw(base_size = 12) +
    theme(
      axis.text.x = element_text(
        angle = 0,
        hjust = 1,
        vjust = 1,
        face = "bold"),
      axis.text.y = element_text(face = "bold"),
      panel.grid = element_blank(),
      legend.position = "bottom",legend.title.position = "bottom",
      legend.text = element_text(size = 7.5, face = "bold"),
      legend.title = element_text(size = 8, face = "bold"))
}
traits <- c("GY", "OHSW", "PN", "PH", "NDF", "NDM")

BLUP_dots <- lapply(traits, function(tr){
  plot_trait_dot(mixedmodel, tr)
})

library(ggpubr)
#BLUP_dots
```

```
blups_d_panel <- ggarrange(
  plotlist = BLUP_dots,
  ncol = 6, nrow = 1,
  labels = c("A", "B", "C", "D", "E", "F"), vjust = 0.11,
  font.label = list(face = "bold"),label.y = 0.01,
  align = "v")
blups_d_panel
```




```
ggsave(
  "BLUPs_d2_LB.pdf",
  blups_d_panel, height = 11, width = 13,
  dpi = 600)
```

Correlation

```
corrplot = plot(
  corr_coef(LB, NDF, NDM, PH, PN, GY, OHSW),
  type = "upper",
  diag = FALSE,
  reorder = T,
  signif = c("stars", "pval"),
  show = c("all", "signif"),
  p_val = 0.05,
  caption = TRUE,
  digits.cor = 2,
  digits.pval = 3,
  col.low = "orange",
  col.mid = "white",
  col.high = "purple",
```

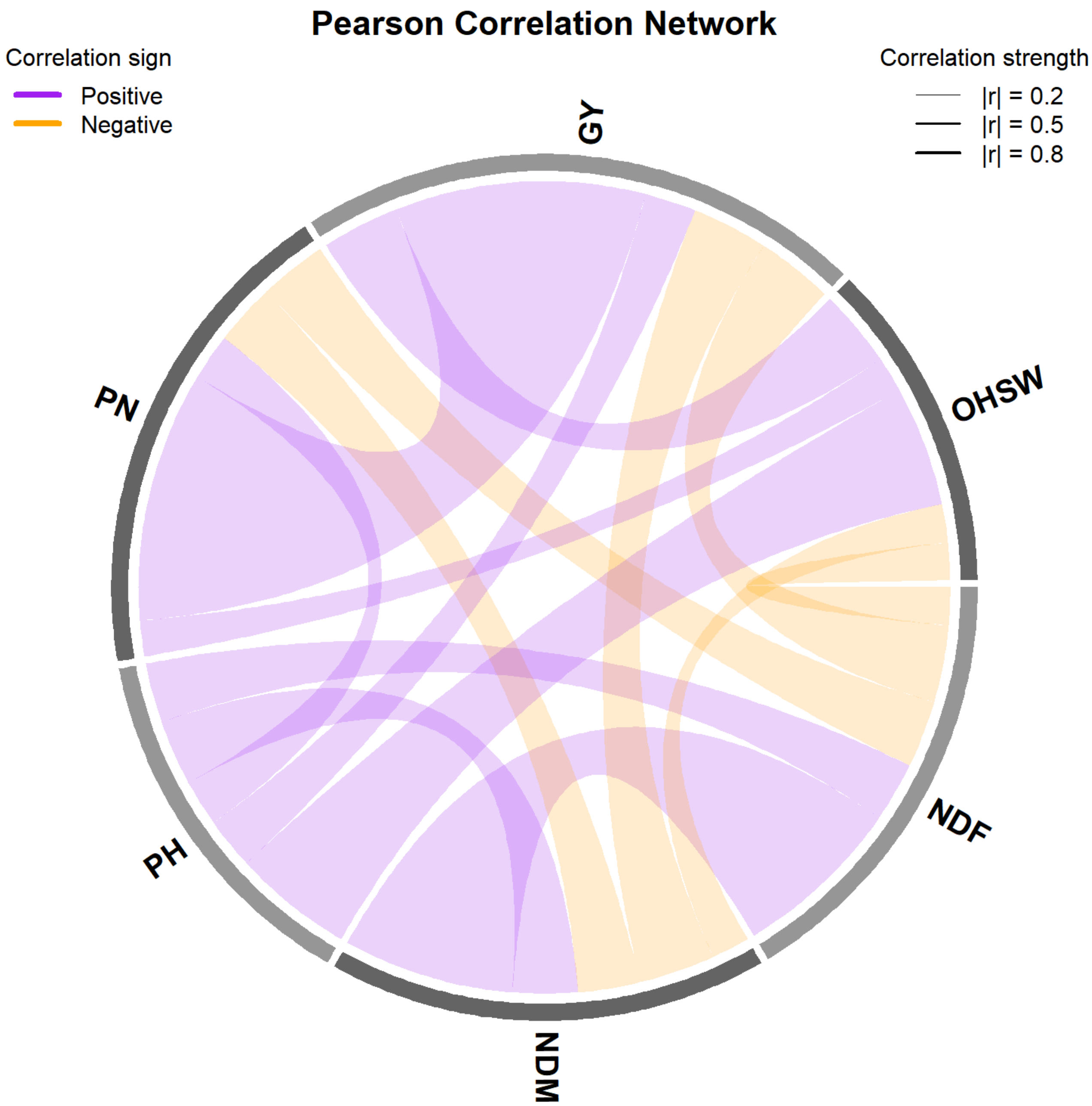
```
  lab.x.position = "top",
  lab.y.position = "left",
  legend.position = "right",
  legend.title = "Pearson's\nCorrelation",
  size.text.cor = 5,
  size.text.signif = 5,
  size.text.lab = 14)
```

```
ggsave("Correlation.pdf",corrplot, width = 7, height = 6, dpi = 600)
```

```
library(circlize)
# Correlation matrix
corr_mat <- matrix(
  c(
    1.000, 0.587, 0.205,-0.230,-0.264,-0.129,
    0.587, 1.000, 0.224,-0.214,-0.262,-0.131,
    0.205, 0.224, 1.000, 0.160, 0.177, 0.413,
   -0.230,-0.214, 0.160, 1.000, 0.861, 0.123,
   -0.264,-0.262, 0.177, 0.861, 1.000, 0.267,
   -0.129,-0.131, 0.413, 0.123, 0.267, 1.000),
  nrow = 6,
  byrow = TRUE)
traits <- c(
  "NDF",
  "NDM",
  "PH",
  "PN",
  "GY",
  "OHSW")
rownames(corr_mat) <- traits
colnames(corr_mat) <- traits
# Convert matrix to links
corr_df <- data.frame()
for(i in 1:(nrow(corr_mat)-1)){
  for(j in (i+1):ncol(corr_mat)){
    corr_df <- rbind(
      corr_df,
      data.frame(
        from  = rownames(corr_mat)[i],
        to    = colnames(corr_mat)[j],
        value = corr_mat[i,j]))
  }
}
# Link colors
corr_df$link_col <- ifelse(
  corr_df$value > 0,
  "purple",
  "orange")
# Trait colors
grid_cols <- c(
  NDF  = "grey60",
  NDM  = "grey40",
  PH   = "grey60",
```

```
  PN   = "grey40",
  GY   = "grey60",
  OHSW = "grey40")
# Plot
circos.clear()
chordDiagram(
  x = corr_df[, c("from","to","value")],
  grid.col = grid_cols,
  col = corr_df$link_col,
  transparency = 0.80,
  directional = 0,
  # MUCH THINNER LINES
  link.lwd = 0.5 + abs(corr_df$value) * 2,
  annotationTrack = "grid",
  preAllocateTracks = list(
    track.height = 0.12))
# Trait labels
circos.track(
  track.index = 1,
  panel.fun = function(x,y){
    circos.text(
      CELL_META$xcenter,
      CELL_META$ylim[1],
      CELL_META$sector.index,
      facing = "clockwise",
      niceFacing = TRUE,
      adj = c(0,0.5),
      font = 2,
      cex = 1.3)
  },
  bg.border = NA
)
# Title
title(
  "Pearson Correlation Network",
  font.main = 2,
  cex.main = 1.4)
# Legends
legend(
  "topleft",
  legend = c(
    "Positive",
    "Negative"),
  col = c(
    "purple",
    "orange"),
  lwd = 4,
  title = "Correlation sign",
  bty = "n",
  cex = 1)
legend(
  "topright",
  legend = c(
    "|r| = 0.2",
```

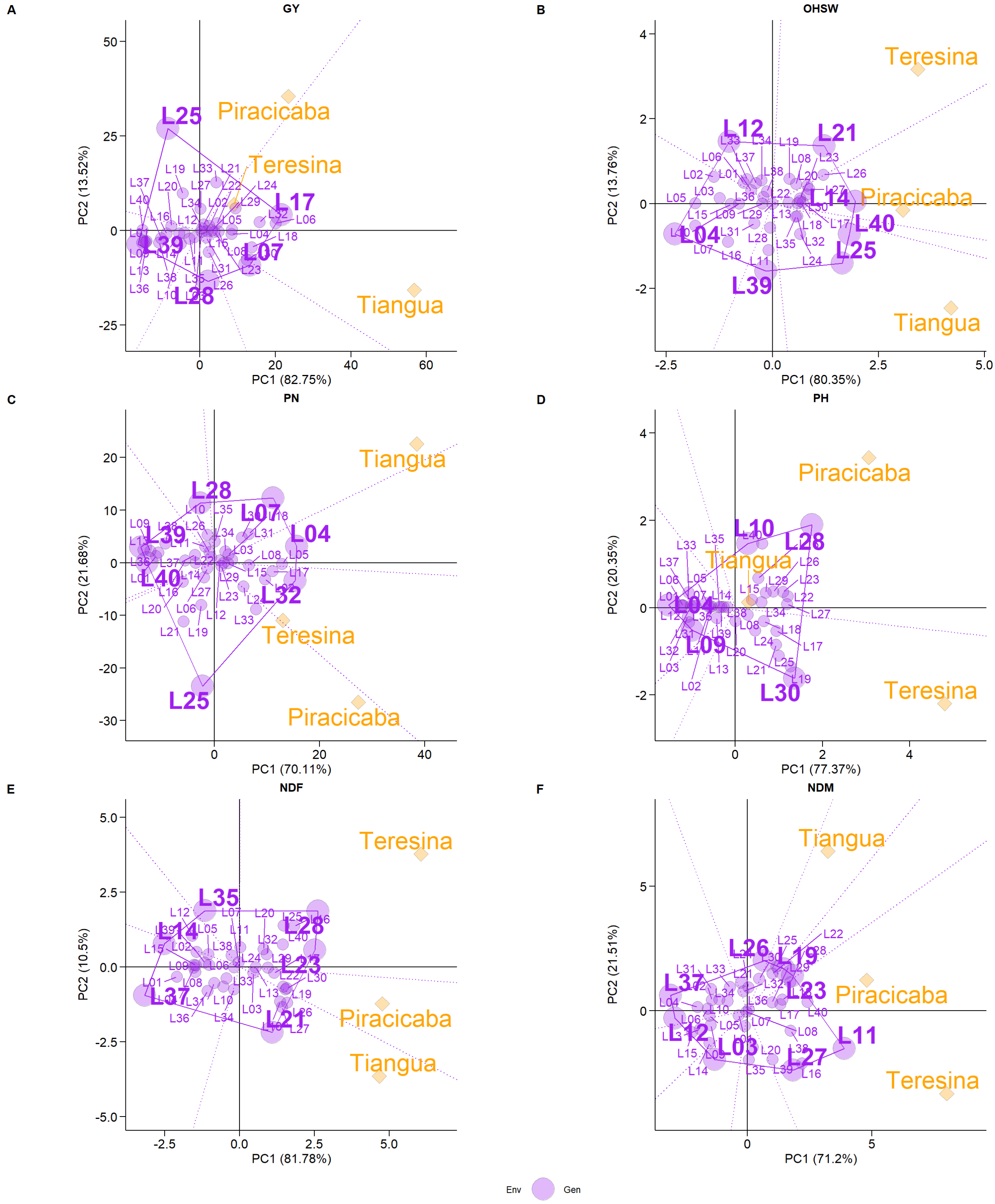
```
    "|r| = 0.5",
    "|r| = 0.8"),
  lwd = c(
    0.5 + 0.2*2,
    0.5 + 0.5*2,
    0.5 + 0.8*2),
  col = "black",
  title = "Correlation strength",
  bty = "n",
  cex = 1)
```




```
pdf(
  "Correlation_Chord_Diagram.pdf",
  width = 8,
  height = 8)
dev.off()
```

#GGE BIPLOT ## Which-Won-where

```
www <- list()

for(tr in traits){
  gge3_tr <- gge(LB, Env, GEN, resp = tr, svp = "symmetrical")

  p <- plot(gge3_tr,
            type = 3,
            col.env = "orange",
            col.gen = "purple",
            shape.gen = 21,
            shape.env = 23,
            col.alpha = 0.3,
            arrow.color = "purple",
            size.text.gen = 4,
            size.text.win = 8,
            size.text.env = 8,
            size.shape = 5,
            size.shape.win = 10,
            large_label = 10,
            repel = TRUE,
            max_overlaps = 40,
            repulsion =8,
            col.line = "orange",
            plot_theme = theme_metan_minimal())

  p_try <- try({
    p2 <- p + labs(title = tr, subtitle = NULL) +
            theme(plot.title = element_text(hjust = 0.5, face = "bold"))
    p <- p2
    NULL
  }, silent = TRUE)

  if(inherits(p_try, "try-error") || !inherits(p, "ggplot")) {
    p_grob <- ggpubr::as_ggplot(p)
    p <- p_grob + labs(title = tr, subtitle = NULL) +
              theme(plot.title = element_text(hjust = 0.5, face = "bold"))
  }

  www[[tr]] <- p
}
www_lb <- ggarrange(
  www[["GY"]],  www[["OHSW"]], www[["PN"]],
  www[["PH"]],  www[["NDF"]],  www[["NDM"]],
  ncol = 2, nrow = 3,
  common.legend = TRUE, legend = "bottom",
  labels = c("A","B","C","D","E","F"),
  font.label = list(size = 12, face = "bold"),
  align = "hv",
  hjust = -0.5,
  vjust = 1.5)
www_lb
```




```r
ggsave("Which-won-Where2.pdf", www_lb,width = 14, height = 16, dpi = 600)
```

Mean_performance vs. stability

```r
dir.create("Mean_performance_vs_stability", showWarnings = FALSE)

mps_list <- list()

for(tr in traits){
    gge2_tr <- gge(
```

```
    LB,
    Env,
    GEN,
    resp = tr,
    svp = "genotype")
  # Plot type 2
  p <- plot(gge2_tr,
            type = 2,
            col.env = "orange",
            col.gen = "purple",
            shape.gen = 21,
            shape.env = 23,
            col.alpha = 0.3,
            arrow.color = "purple",
            size.text.gen = 4,
            size.text.env = 5,
            size.shape = 5,
            large_label = 1,
            repel = TRUE,
            max_overlaps = 10,
            repulsion =40,
            col.line = "orange",
            axis_expand = 1.2,
            plot_theme = theme_metan_minimal())

  p_try <- try({
    p2 <- p + labs(title = tr, subtitle = NULL) +
      theme(plot.title = element_text(hjust = 0.5, face = "bold"))
    p <- p2
    NULL
  }, silent = TRUE)

  if(inherits(p_try, "try-error") || !inherits(p, "ggplot")) {
    p_grob <- ggpubr::as_ggplot(p)
    p <- p_grob + labs(title = tr, subtitle = NULL) +
      theme(plot.title = element_text(hjust = 0.5, face = "bold"))
  }

  mps_list[[tr]] <- p2}

mps_panel <- ggarrange(
  mps_list[["GY"]],  mps_list[["OHSW"]], mps_list[["PN"]],
  mps_list[["PH"]],  mps_list[["NDF"]],  mps_list[["NDM"]],
  ncol = 2, nrow = 3,
  common.legend = TRUE, legend = "bottom",
  labels = c("A","B","C","D","E","F"),
  font.label = list(size = 10, face = "bold"),
  align = "hv")
mps_panel = annotate_figure(
  mps_panel,
  top = text_grob("", size = 14),
  fig.lab = "",
  fig.lab.pos = "bottom")
```

```
mps_GY <- gge(
  LB,
  Env,
  GEN,
  resp = "GY",
  svp = "genotype")

mpsp_GY <- plot(
  mps_GY,
  type = 2,
  col.env = "orange",
  col.gen = "purple",
  col.line = "grey50",
  size.text.gen = 4,
  size.text.env = 5,
  repel = TRUE,
  repulsion = 0,
  max_overlaps = 40,
  axis_expand = 1.1,
  plot_theme = theme_classic(base_size = 12)) +
  labs(title = NULL, subtitle = NULL)

mpsp_OHSW <- gge(
  LB,
  Env,
  GEN,
  resp = "OHSW",
  svp = "genotype")

mpspp_OHSW <- plot(
  mpsp_OHSW,
  type = 2,
  col.env = "orange",
  col.gen = "purple",
  col.line = "grey50",
  size.text.gen = 4,
  size.text.env = 5,
  repel = TRUE,
  repulsion = 20,
  max_overlaps = 30,
  axis_expand = 1.08,
  plot_theme = theme_classic(base_size = 14)) +
  labs(title = NULL, subtitle = NULL)

mpsp_PN <- gge(
  LB,
  Env,
  GEN,
  resp = "PN",
  svp = "genotype")

mpspp_PN <- plot(
  mpsp_PN,
  type = 2,
```

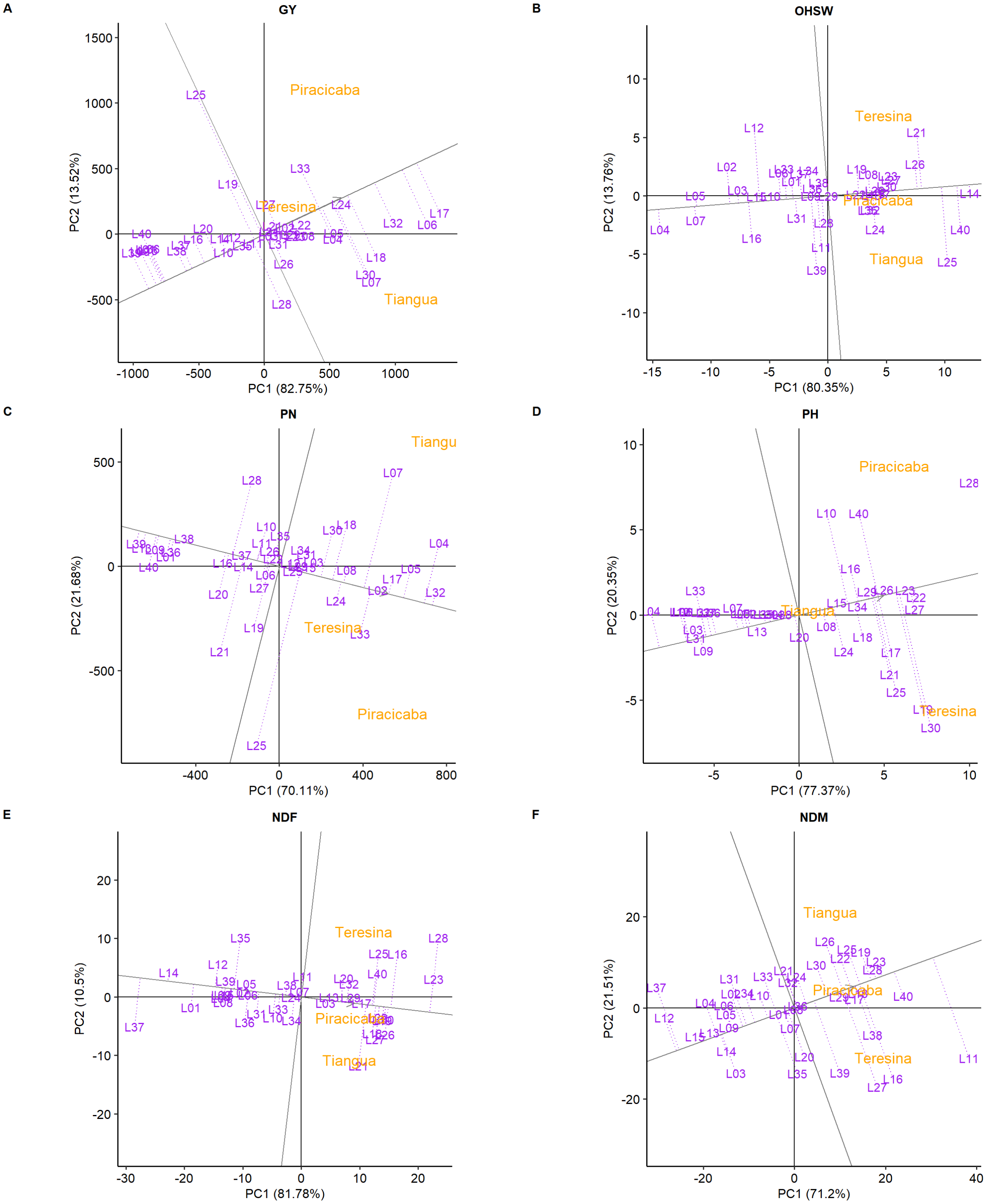
```
  col.env = "orange",
  col.gen = "purple",
  col.line = "grey50",
  size.text.gen = 4,
  size.text.env = 5,
  repel = TRUE,
  repulsion = 20,
  max_overlaps = 30,
  axis_expand = 1.1,
  plot_theme = theme_classic(base_size = 14)) +
  labs(title = NULL, subtitle = NULL)

mpsp_PH <- gge(
  LB,
  Env,
  GEN,
  resp = "PH",
  svp = "genotype")

mpspp_PH <- plot(
  mpsp_PH,
  type = 2,
  col.env = "orange",
  col.gen = "purple",
  col.line = "grey50",
  size.text.gen = 4,
  size.text.env = 5,
  repel = TRUE,
  repulsion = 20,
  max_overlaps = 30,
  axis_expand = 1.05,
  plot_theme = theme_classic(base_size = 14)) +
  labs(title = NULL, subtitle = NULL)

mpsp_NDF <- gge(
  LB,
  Env,
  GEN,
  resp = "NDF",
  svp = "genotype")

mpspp_NDF <- plot(
  mpsp_NDF,
  type = 2,
  col.env = "orange",
  col.gen = "purple",
  col.line = "grey50",
  size.text.gen = 4,
  size.text.env = 5,
  repel = TRUE,
  repulsion = 20,
  max_overlaps = 30,
  axis_expand = 1.1,
  plot_theme = theme_classic(base_size = 14)) +
```

```
  labs(title = NULL, subtitle = NULL)

mpsp_NDM <- gge(
  LB,
  Env,
  GEN,
  resp = "NDM",
  svp = "genotype")

mpspp_NDM <- plot(
  mpsp_NDM,
  type = 2,
  col.env = "orange",
  col.gen = "purple",
  col.line = "grey50",
  size.text.gen = 4,
  size.text.env = 5,
  repel = TRUE,
  repulsion = 20,
  max_overlaps = 30,
  axis_expand = 1.07,
  plot_theme = theme_classic(base_size = 14)) +
  labs(title = NULL, subtitle = NULL)
```

```
mpsp_GY <- mpsp_GY +
  ggtitle("GY") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

mpspp_OHSW <- mpspp_OHSW +
  ggtitle("OHSW") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

mpspp_PN <- mpspp_PN +
  ggtitle("PN") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

mpspp_PH <- mpspp_PH +
  ggtitle("PH") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

mpspp_NDF <- mpspp_NDF +
  ggtitle("NDF") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

mpspp_NDM <- mpspp_NDM +
  ggtitle("NDM") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```

```
common_theme <- theme(
  legend.position = "bottom",
  panel.grid = element_blank(),
  plot.title = element_text(
    hjust = 0.5,
```

```
    face = "bold",
    size = 12
  )
)
mpsp_GY     <- mpsp_GY + common_theme
mpspp_OHSW  <- mpspp_OHSW + common_theme
mpspp_PN    <- mpspp_PN + common_theme
mpspp_PH    <- mpspp_PH + common_theme
mpspp_NDF   <- mpspp_NDF + common_theme
mpspp_NDM   <- mpspp_NDM + common_theme
mps_panel <- ggarrange(
  mpsp_GY,
  mpspp_OHSW,
  mpspp_PN,
  mpspp_PH,
  mpspp_NDF,
  mpspp_NDM,
  ncol = 2,
  nrow = 3,
  common.legend = TRUE,
  legend = "bottom",
  labels = c("A","B","C","D","E","F"),
  font.label = list(
    size = 12,
    face = "bold"
  )
)
mps_panel
```




```r
ggsave("Mean Performance vs Stability_LB2.pdf", mps_panel, width = 14, height = 16, dpi = 600)
```

Discriminativeness vs. representativeness

```r
library(ggplot2)
library(ggpubr)

dir.create("Discriminativeness_vs_Representativeness", showWarnings = FALSE)

dr_list <- list()
```

```
for(tr in traits){

  gge4_tr <- gge(
    LB,
    Env,
    GEN,
    resp = tr,
    centering = "environment",
    scaling = "none",
    svp = "environment"
  )

  p4 <- plot(gge4_tr,
             type = 4,
             col.env = "orange",
             col.gen = "purple",
             size.text.gen = 4,
             size.text.env = 4 ,
             repel = TRUE,
             repulsion = 10,
             col.line = "orange",
             size.line = 0.5,
             col.circle = "grey",
             col.alpha = 0.4,
             max_overlaps = 20,
             plot_theme = theme_metan_minimal())
  p_try <- try({
    p4_mod <- p4 +
      labs(title = tr, subtitle = NULL) +
      theme(plot.title = element_text(hjust = 0.5, face = "bold"))
    p4 <- p4_mod
    NULL
  }, silent = TRUE)
  if(inherits(p_try, "try-error") || !inherits(p4, "ggplot")) {
    p_grob <- ggpubr::as_ggplot(p4)
    p4 <- p_grob +
      labs(title = tr, subtitle = NULL) +
      theme(plot.title = element_text(hjust = 0.5, face = "bold"))
  }
  dr_list[[tr]] <- p4
}
dr_panel <- ggarrange(
  dr_list[["GY"]],  dr_list[["OHSW"]], dr_list[["PN"]],
  dr_list[["PH"]],  dr_list[["NDF"]],  dr_list[["NDM"]],
  ncol = 2, nrow = 3,
  common.legend = TRUE, legend = "bottom",
  labels = c("A","B","C","D","E","F"),
  font.label = list(size = 10, face = "bold"),
  align = "hv",
  hjust = -0.5,
  vjust = 1.5
)
dr_panel = annotate_figure(
  dr_panel,
```

```
  top = text_grob("", size = 14),
  fig.lab = "",
  fig.lab.pos = "bottom"
)
```

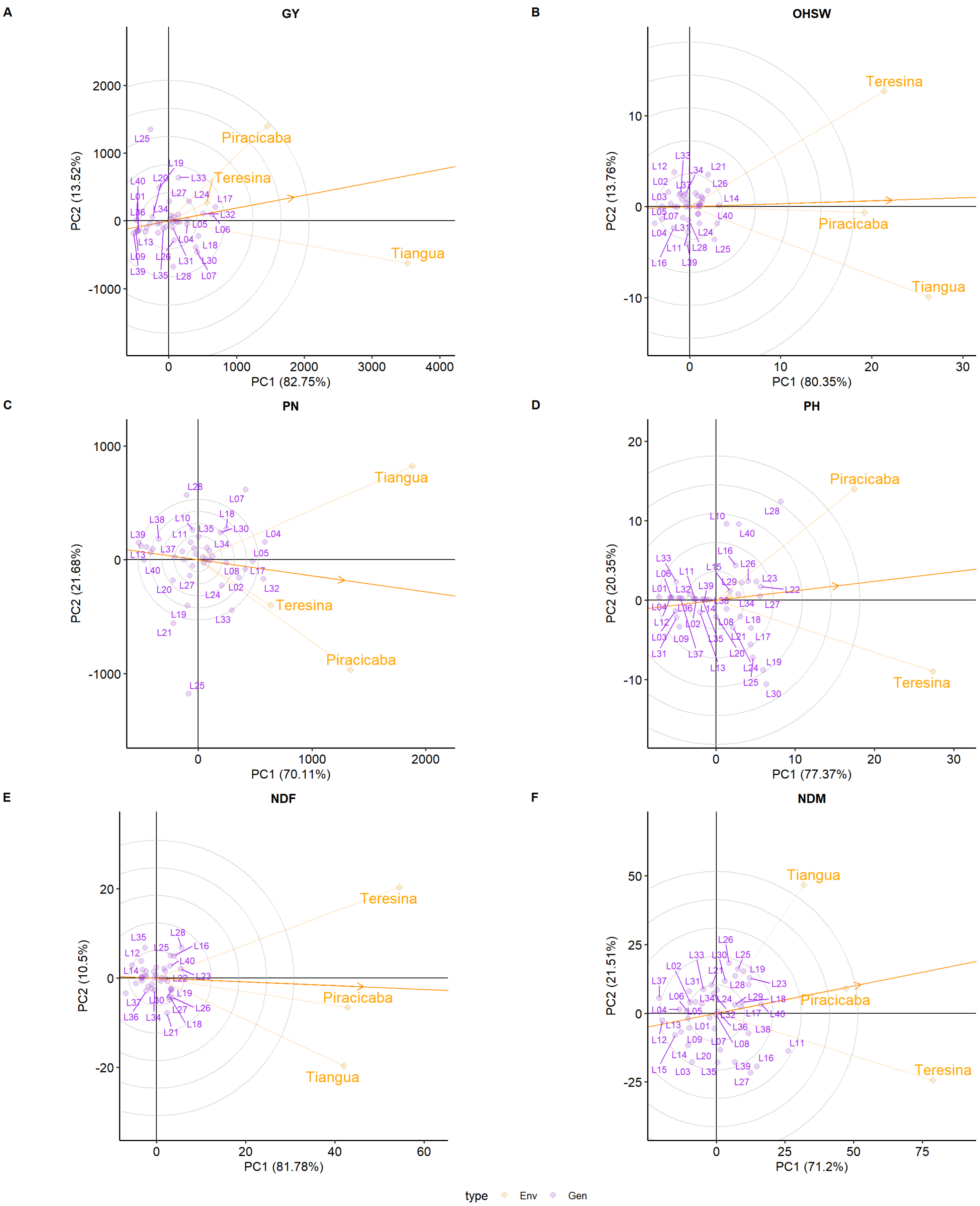
```
dr_GY <- gge(
  LB,
  Env,
  GEN,
  resp = "GY",
  centering = "environment",
  scaling = "none",
  svp = "environment"
)

drp_GY <- plot(
  dr_GY,
  type = 4,
  col.env = "orange",
  col.gen = "purple",
  size.text.gen = 3,
  size.text.env = 5,
  col.alpha = 0.2,
  repel = TRUE,
  repulsion = 5,
  max_overlaps = 20,
  col.line = "darkorange",
  plot_theme = theme_classic(base_size = 12)
) +
  labs(title = NULL, subtitle = NULL)

dr_OHSW <- gge(
  LB,
  Env,
  GEN,
  resp = "OHSW",
  centering = "environment",
  scaling = "none",
  svp = "environment"
)

drp_OHSW <- plot(
  dr_OHSW,
  type = 4,
  col.env = "orange",
  col.gen = "purple",
  size.text.gen = 3,
  size.text.env = 5,
  col.alpha = 0.2,
  repel = TRUE,
  repulsion = 5,
  max_overlaps = 20,
  col.line = "darkorange",
```

```
  plot_theme = theme_classic(base_size = 12)
) +
  labs(title = NULL, subtitle = NULL)

dr_PN <- gge(
  LB,
  Env,
  GEN,
  resp = "PN",
  centering = "environment",
  scaling = "none",
  svp = "environment"
)

drp_PN <- plot(
  dr_PN,
  type = 4,
  col.env = "orange",
  col.gen = "purple",
  size.text.gen = 3,
  size.text.env = 5,
  col.alpha = 0.2,
  repel = TRUE,
  repulsion = 5,
  max_overlaps = 10,
  col.line = "darkorange",
  plot_theme = theme_classic(base_size = 12)
) +
  labs(title = NULL, subtitle = NULL)

dr_PH<- gge(
  LB,
  Env,
  GEN,
  resp = "PH",
  centering = "environment",
  scaling = "none",
  svp = "environment"
)

drp_PH <- plot(
  dr_PH,
  type = 4,
  col.env = "orange",
  col.gen = "purple",
  size.text.gen = 3,
  size.text.env = 5,
  col.alpha = 0.2,
  repel = TRUE,
  repulsion = 5,
  max_overlaps = 20,
  col.line = "darkorange",
  plot_theme = theme_classic(base_size = 12)
) +
```

```
  labs(title = NULL, subtitle = NULL)

dr_NDF<- gge(
  LB,
  Env,
  GEN,
  resp = "NDF",
  centering = "environment",
  scaling = "none",
  svp = "environment"
)

drp_NDF <- plot(
  dr_NDF,
  type = 4,
  col.env = "orange",
  col.gen = "purple",
  size.text.gen = 3,
  size.text.env = 5,
  col.alpha = 0.2,
  repel = TRUE,
  repulsion = 5,
  max_overlaps = 20,
  col.line = "darkorange",
  plot_theme = theme_classic(base_size = 12)
) +
  labs(title = NULL, subtitle = NULL)

dr_NDM<- gge(
  LB,
  Env,
  GEN,
  resp = "NDM",
  centering = "environment",
  scaling = "none",
  svp = "environment"
)

drp_NDM <- plot(
  dr_NDM,
  type = 4,
  col.env = "orange",
  col.gen = "purple",
  size.text.gen = 3,
  size.text.env = 5,
  col.alpha = 0.2,
  repel = TRUE,
  repulsion = 5,
  max_overlaps = 10,
  col.line = "darkorange",
  plot_theme = theme_classic(base_size = 12)
) +
  labs(title = NULL, subtitle = NULL)
```

```
drp_GY <- drp_GY +
  ggtitle("GY") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

drp_OHSW <- drp_OHSW +
  ggtitle("OHSW") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

drp_PN <- drp_PN +
  ggtitle("PN") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

drp_PH <- drp_PH +
  ggtitle("PH") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

drp_NDF <- drp_NDF +
  ggtitle("NDF") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

drp_NDM <- drp_NDM +
  ggtitle("NDM") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
common_theme <- theme(
  legend.position = "bottom",
  panel.grid = element_blank(),
  plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 12))
drp_GY    <- drp_GY    + common_theme
drp_OHSW  <- drp_OHSW  + common_theme
drp_PN    <- drp_PN    + common_theme
drp_PH    <- drp_PH    + common_theme
drp_NDF   <- drp_NDF   + common_theme
drp_NDM   <- drp_NDM   + common_theme
dr_panel <- ggarrange(
  drp_GY,
  drp_OHSW,
  drp_PN,
  drp_PH,
  drp_NDF,
  drp_NDM,
  ncol = 2,
  nrow = 3,
  common.legend = TRUE,
  legend = "bottom",
  labels = c("A","B","C","D","E","F"),
  font.label = list(
    size = 12,
    face = "bold"
  )
```

```
)
dr_panel
```




```
ggsave("Discriminativeness vs Representativeness_LB.pdf", dr_panel, width = 14, height = 16, d
```

Simultaneous Selection

Desire Gain Index

```
# DESIRE INDEX FUNCTION
gainDiff <- function(ebvMat, b_ebv, dGain, nSel){
  take <- order(ebvMat %*% b_ebv, decreasing = TRUE)[1:nSel]
  mu_Sel <- colMeans(ebvMat[take, ])
  resp   <- mu_Sel - colMeans(ebvMat)
  resp   <- resp / sqrt(sum(resp^2))
  sum((resp - dGain)^2)
}
```

```
ebvMat <- ebvMat[complete.cases(ebvMat), ]
#COV MATRIX -EBVS
library(AlphaSimR)
```

```
Warning: pacote 'AlphaSimR' foi compilado no R versão 4.4.3

Carregando pacotes exigidos: R6


Anexando pacote: 'AlphaSimR'

O seguinte objeto é mascarado por 'package:ggpubr':

    mutate

O seguinte objeto é mascarado por 'package:metan':

    mutate

O seguinte objeto é mascarado por 'package:dplyr':

    mutate
```

```
(G_ebv <- AlphaSimR::popVar(ebvMat))
```

```
           [,1]       [,2]       [,3]       [,4]      [,5]       [,6]
[1,] 6619.90441 218.241188 423.564858 252.987379 34.774836 4066.91625
[2,]  218.24119   7.357991  14.144528   8.392790  1.154951  134.17944
[3,]  423.56486  14.144528  28.506912  16.749936  2.294614  263.66338
[4,]  252.98738   8.392790  16.749936  10.010313  1.366469  156.84246
[5,]   34.77484   1.154951   2.294614   1.366469  0.190287   21.55917
[6,] 4066.91625 134.179444 263.663377 156.842455 21.559165 2535.04293
```

```
#desired gains
desiredGain <- c(
  GY   =  2,
  OHSW =  1,
  NDF  = -1,
  NDM  = -1,
  PH   =  0,
  PN   =  1
)
```

```
desiredGain <- desiredGain / sqrt(sum(desiredGain^2))
sum(desiredGain^2)
```

```
[1] 1
```

```
b_ebv <- solve(G_ebv) %*% desiredGain
```

```
nSel <- 5
opt <- optim(
  par     = b_ebv,
  fn      = gainDiff,
  ebvMat  = ebvMat,
  dGain   = desiredGain,
  nSel    = nSel
)
(opt_weights <- opt$par)
```

```
            [,1]
[1,] -0.06310990
[2,]  2.58214590
[3,] -0.31518794
[4,] -2.20499579
[5,]  8.86931984
[6,]  0.05848843
```

```
DesireIndex <- ebvMat %*% opt_weights
DesireIndex <- data.frame(
  GEN = rownames(ebvMat),
  DesireIndex = as.numeric(DesireIndex)) |>
  arrange(desc(DesireIndex))
head(DesireIndex, 10)
```

```
   GEN DesireIndex
1  L27    242.0702
2  L20    241.6511
3  L21    241.6220
4  L13    241.4399
5  L06    241.3781
6  L34    241.0957
7  L14    240.9096
8  L11    240.7632
9  L12    240.4101
10 L25    240.2991
```

```
write.xlsx(DesireIndex, "Desireindex.xlsx")
```

```
(selected <- DesireIndex[1:5, ])
```

```
  GEN DesireIndex
1 L27    242.0702
2 L20    241.6511
```

```
3 L21     241.6220
4 L13     241.4399
5 L06     241.3781
```

```
sel_mat <- ebvMat[selected$GEN, ]

expResp <- colMeans(sel_mat) - colMeans(ebvMat)
scaleExpResp <- expResp / sqrt(sum(expResp^2))

df_vect = cbind(
  Desired  = desiredGain,
  Achieved = scaleExpResp)
```

```
library(dplyr)
df <- dplyr::arrange(DesireIndex, dplyr::desc(DesireIndex)) |>
  dplyr::mutate(
    GEN = factor(GEN, levels = GEN),
    Selected = ifelse(GEN %in% selected$GEN, "Selected", "Not selected"))
```

```
library(ggplot2)

desiredgainplot = ggplot(df, aes(x = GEN, y = DesireIndex, fill = Selected)) +
  geom_col(width = 0.9, color = "black", size = 0.2) +
  coord_polar() +
  scale_fill_manual(
    values = c(
      "Selected"     = "#6A1B9A",
      "Not selected" = "orange")) +
  labs(face = "bold",
    fill  = NULL) +
  theme_minimal(base_size = 14) +
  theme(
    axis.text.x = element_text(face = "bold"),
    axis.title = element_blank(),
    axis.text.y = element_blank(),
    axis.ticks = element_blank(),
    panel.grid = element_blank(),
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "bottom")
```

```
ggsave("desiredgainplot_LB.pdf", plot = desiredgainplot,width = 10, height = 8,dpi = 600)
```

```
h2_df <- gmd(mixedmodel, "h2")
```

```
Class of the model: waasb

Variable extracted: h2
```

```
h2 <- h2_df$h2
names(h2) <- h2_df$VAR
traits_DI <- colnames(ebvMat)
```

```r
h2_DI <- h2[traits_DI]
library(dplyr)

genetic_gain_table <- function(ebvMat, selected_gen, h2){

  traits <- colnames(ebvMat)

  res <- lapply(traits, function(tr){

    Xo <- mean(ebvMat[, tr], na.rm = TRUE)
    Xs <- mean(ebvMat[selected_gen, tr], na.rm = TRUE)

    SD  <- Xs - Xo
    SDp <- (SD / Xo) * 100

    SG  <- SD * h2[tr]
    SGp <- (SG / Xo) * 100

    tibble(
      Trait    = tr,
      `h²` = round(h2[tr], 2),
      Xo       = round(Xo, 2),
      Xs       = round(Xs, 2),
      SD       = round(SD, 2),
      `SD (%)` = round(SDp, 2),
      SG       = round(SG, 2),
      `SG (%)` = round(SGp, 2)
    )
  })

  bind_rows(res)
}
Table3_DesireIndex <- genetic_gain_table(
  ebvMat = ebvMat,
  selected_gen = selected$GEN,
  h2 = h2_DI)
```

```r
h2_cullis <- Cullis_all$Cullis_H2
names(h2_cullis) <- Cullis_all$Trait
h2_cullis
```

```
       PH       NDF       NDM      OHSW        PN        GY 
0.1824919 0.7520623 0.4029824 0.7543363 0.2807144 0.3034326 
```

```r
Table3_DesireIndex_Cullis <- genetic_gain_table(
  ebvMat = ebvMat,
  selected_gen = selected$GEN,
  h2 = h2_cullis)
Table3_DesireIndex_Cullis
```

```
# A tibble: 6 × 8
  Trait  `h²`    Xo    Xs    SD `SD (%)`    SG `SG (%)`
  <chr> <dbl> <dbl> <dbl> <dbl>    <dbl> <dbl>    <dbl>
```

```
1 GY     0.3  749.  766.  17.7      2.36 5.36     0.72
2 OHSW   0.75  32.2  33.2  1.01     3.12 0.76     2.36
3 NDF    0.75  43.6  44.5  0.91     2.08 0.68     1.56
4 NDM    0.4   87.8  88.2  0.39     0.44 0.16     0.18
5 PH     0.18  42.5  42.7  0.14     0.34 0.03     0.06
6 PN     0.28 575.  586.  11.1      1.94 3.13     0.54
```

```
Table_Broad <- genetic_gain_table(
  ebvMat,
  selected$GEN,
  h2_DI) %>%
  dplyr::rename(
    H2_Broad = `h²`,
    SG_Broad = SG,
    `SG_Broad (%)` = `SG (%)`)

Table_Cullis <- genetic_gain_table(
  ebvMat,
  selected$GEN,
  h2_cullis) %>%
  dplyr::rename(
    H2_Cullis = `h²`,
    SG_Cullis = SG,
    `SG_Cullis (%)` = `SG (%)`)

CompareGain <- Table_Broad %>%
  dplyr::select(Trait, H2_Broad, SG_Broad, `SG_Broad (%)`) %>%
  left_join(
    Table_Cullis %>%
      dplyr::select(Trait, H2_Cullis, SG_Cullis, `SG_Cullis (%)`),
    by = "Trait")
CompareGain
```

```
# A tibble: 6 × 7
  Trait H2_Broad SG_Broad `SG_Broad (%)` H2_Cullis SG_Cullis `SG_Cullis (%)`
  <chr>    <dbl>    <dbl>          <dbl>     <dbl>     <dbl>           <dbl>
1 GY        0.36     6.36           0.85      0.3       5.36            0.72
2 OHSW      0.81     0.81           2.52      0.75      0.76            2.36
3 NDF       0.8      0.73           1.67      0.75      0.68            1.56
4 NDM       0.47     0.18           0.21      0.4       0.16            0.18
5 PH        0.22     0.03           0.08      0.18      0.03            0.06
6 PN        0.33     3.73           0.65      0.28      3.13            0.54
```

```
library(knitr)
```

```
Warning: pacote 'knitr' foi compilado no R versão 4.4.3
```

```
tableDesire =kable(Table3_DesireIndex)
```

```
write.xlsx(Table3_DesireIndex, "Table3_DesireIndex.xlsx", rowNames = FALSE)
```

Populations

```r
library(tibble)
populations <- tribble(
  ~Population, ~Cross, ~Male_origin, ~Female_origin, ~Derived_lines,
  "H25", "G25236 × BGP UFPI 628",
  "Buenos Aires, Argentina (CIAT)",
  "Piauí, Brazil (BGP-UFPI)",
  "L01–L16",

  "H46", "UC HASKELL × BGP UFPI 728",
  "California, USA (UC Davis)",
  "Piauí, Brazil (BGP-UFPI)",
  "L17–L30",

  "H50", "BGP UFPI 728 × BGP UFPI 628",
  "Piauí, Brazil (BGP-UFPI)",
  "Piauí, Brazil (BGP-UFPI)",
  "L31–L34",

  "H81", "BGP UFPI 628 × UC 92",
  "Piauí, Brazil (BGP-UFPI)",
  "California, USA (UC Davis)",
  "L35–L37",

  "H94", "BGP UFPI 728 × UC 92",
  "Piauí, Brazil (BGP-UFPI)",
  "California, USA (UC Davis)",
  "L38–L40")
populations
```

```
# A tibble: 5 × 5
  Population Cross                       Male_origin Female_origin Derived_lines
  <chr>      <chr>                       <chr>       <chr>         <chr>
1 H25        G25236 × BGP UFPI 628       Buenos Air… Piauí, Brazi… L01–L16
2 H46        UC HASKELL × BGP UFPI 728   California… Piauí, Brazi… L17–L30
3 H50        BGP UFPI 728 × BGP UFPI 628 Piauí, Bra… Piauí, Brazi… L31–L34
4 H81        BGP UFPI 628 × UC 92        Piauí, Bra… California, … L35–L37
5 H94        BGP UFPI 728 × UC 92        Piauí, Bra… California, … L38–L40
```

```r
library(dplyr)
library(tidyr)
library(stringr)
```

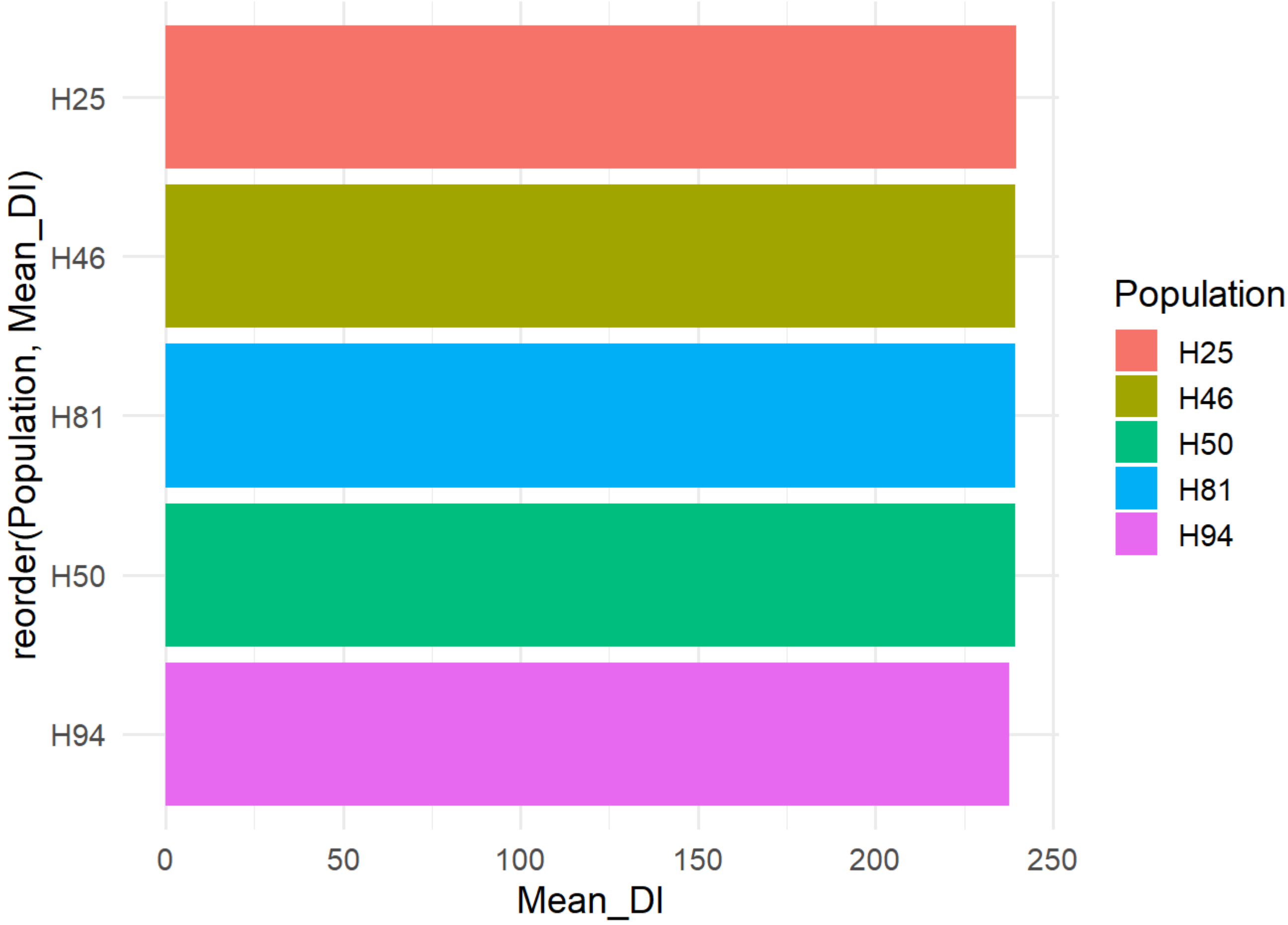
```
Warning: pacote 'stringr' foi compilado no R versão 4.4.3
```

```r
library(purrr)
pop_lines <- populations %>%
  rowwise() %>%
  dplyr::mutate(
    GEN = list(
      sprintf(
```

```
        "L%02d",
        seq(
          as.numeric(str_extract(Derived_lines, "(?<=L)\\d+")),
          as.numeric(str_extract(Derived_lines, "(?<=-L)\\d+"))
        )
      )
    )
  ) %>%
  unnest(GEN) %>%
  ungroup()
```

```
DesireIndex_pop <- DesireIndex %>%
  left_join(
    pop_lines %>%
      dplyr::select(GEN, Population, Cross, Derived_lines),
    by = "GEN")
```

```
pop_summary <- DesireIndex_pop %>%
  group_by(Population, Cross) %>%
  summarise(
    Mean_DI = mean(DesireIndex),
    .groups = "drop")

ggplot(
  pop_summary,
  aes(
    x = reorder(Population, Mean_DI),
    y = Mean_DI,
    fill = Population)) +
  geom_col() +
  coord_flip() +
  theme_minimal(base_size = 14)
```

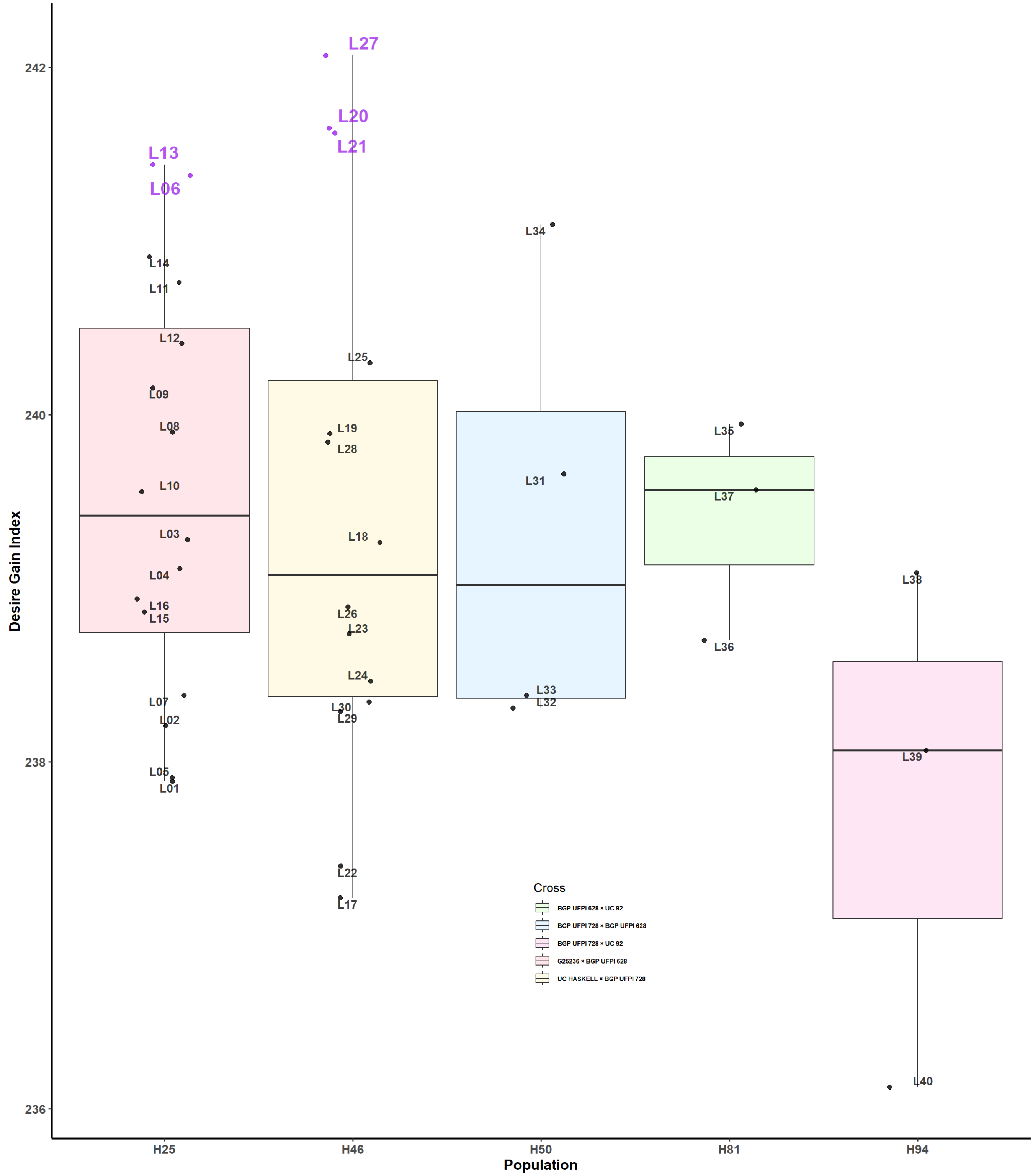



```
top5 <- selected$GEN
DesireIndex_pop <- DesireIndex_pop %>%
  dplyr::mutate(
    Elite = ifelse(GEN %in% top5, "Selected", "Not Selected"))
```

```
library(ggplot2)
library(ggrepel)

DGIPLOT = ggplot(
  DesireIndex_pop,
  aes(
    x = Population,
    y = DesireIndex
  )
) +

  geom_boxplot(
    aes(fill = Cross),
    width = 0.9,
    alpha = 0.13,
    linewidth = 0.5,
    outlier.shape = NA
  ) +

  # Points
  geom_jitter(
```

```
    aes(
      color = Elite,
      shape = Elite),
    width = 0.15,
    size = 3,
    alpha = 0.8) +
  # Labels for non-selected lines
  geom_text_repel(
    data = subset(DesireIndex_pop, Elite == "Not Selected"),
    aes(label = GEN),
    size = 4,
    fontface = "bold",
    alpha = 0.75,
    color = "black",
    max.overlaps = Inf,
    box.padding = 0,
    point.padding = 1,
    segment.alpha = 0) +
  # Labels for selected lines
  geom_text_repel(
    data = subset(DesireIndex_pop, Elite == "Selected"),
    aes(label = GEN),
    size = 6,
    fontface = "bold",
    alpha = 0.75,
    color = "purple",
    max.overlaps = Inf,
    box.padding = 0.3,
    point.padding = 1,
    segment.alpha = 0) +
  scale_shape_manual(
    values = c(
      "Not Selected" = 20,
      "Selected" = 20
    ),
  guide = "none") +
  scale_color_manual(
    values = c(
      "Not Selected" = "black",
      "Selected" = "purple"),
  guide = "none")  +
  scale_fill_manual(
  values = c(
    "G25236 × BGP UFPI 628"       = "#FF5E66",
    "UC HASKELL × BGP UFPI 728"   = "#FFF75E",
    "BGP UFPI 728 × BGP UFPI 628" = "#5EB7FF",
    "BGP UFPI 628 × UC 92"        = "#66FF5E",
    "BGP UFPI 728 × UC 92"        = "#FF5EB7"
  ),
  name = "Cross" )+
  labs(
    x = "Population",
    y = "Desire Gain Index",
    color = NULL,
```

```
    shape = NULL) +
  theme_classic(base_size = 12) +
  theme(
    panel.grid = element_blank(),
    axis.line = element_line(
      colour = "black",
      linewidth = 1),
    axis.title.x = element_text(
      face = "bold",
      size = 14),
    axis.title.y = element_text(
      face = "bold",
      size = 14),
    axis.text.x = element_text(
      face = "bold",
      size = 12),
    axis.text.y = element_text(
      face = "bold",
      size = 12),
    legend.text = element_text(
      face = "bold",
      size = 6),
    legend.position = c(0.55, 0.18)
  )
DGIPLOT
```



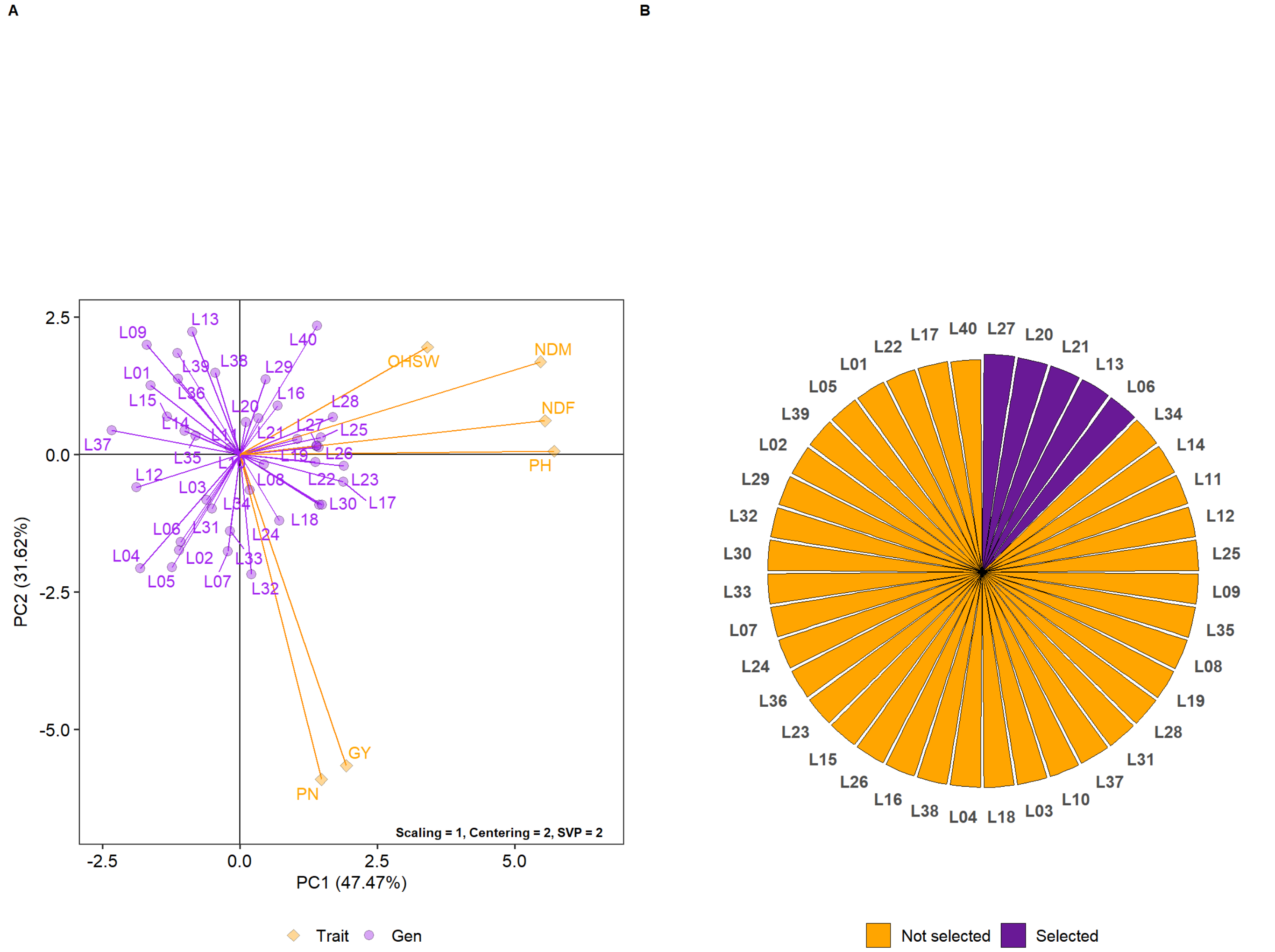
Genotype by trait (GT) biplot

```
library(ggplot2)
gtb1 <- gtb(
  LB,
  GEN,
  resp = c(NDF, NDM, PH, PN, GY, OHSW))
gtb1plot <- plot(
  gtb1,
  col.env = "orange",
```

```
  col.gen = "purple",
  size.text.gen = 4,
  size.text.env = 4,
  repel = TRUE,
  repulsion = 4,
  col.line = "darkorange",
  print.sub = FALSE,
  plot_theme = theme_bw(base_size = 14) +
    theme(
      plot.title = element_blank(),
      plot.subtitle = element_blank(),
      panel.grid = element_blank(),
      legend.position = c(0.85, 0.2),
      legend.title = element_blank()))
gtb1plot$layers[[5]]$aes_params$alpha <- 0.4
gtb1plot$layers[[5]]$aes_params$size  <- 3
gtb1plot <- gtb1plot +
  annotate(
    "text",
    x = Inf,
    y = -Inf,
    label = "Scaling = 1, Centering = 2, SVP = 2",
    hjust = 1.1,
    vjust = -0.8,
    fontface = "bold",
    size = 3
  )
```

Arrange

```
gt_desired_panel <- ggarrange(gtb1plot, desiredgainplot,
  ncol = 2,
  common.legend = F, legend = "bottom",
  labels = c("A","B"),
  font.label = list(size = 10, face = "bold"),
  align = "hv",
  hjust = -0.5,
  vjust = 1.5
)
gt_desired_panel
```




```
ggsave("gt_desired_panel_LB.pdf", plot = gt_desired_panel, dpi = 600)
```

Selection shift plot

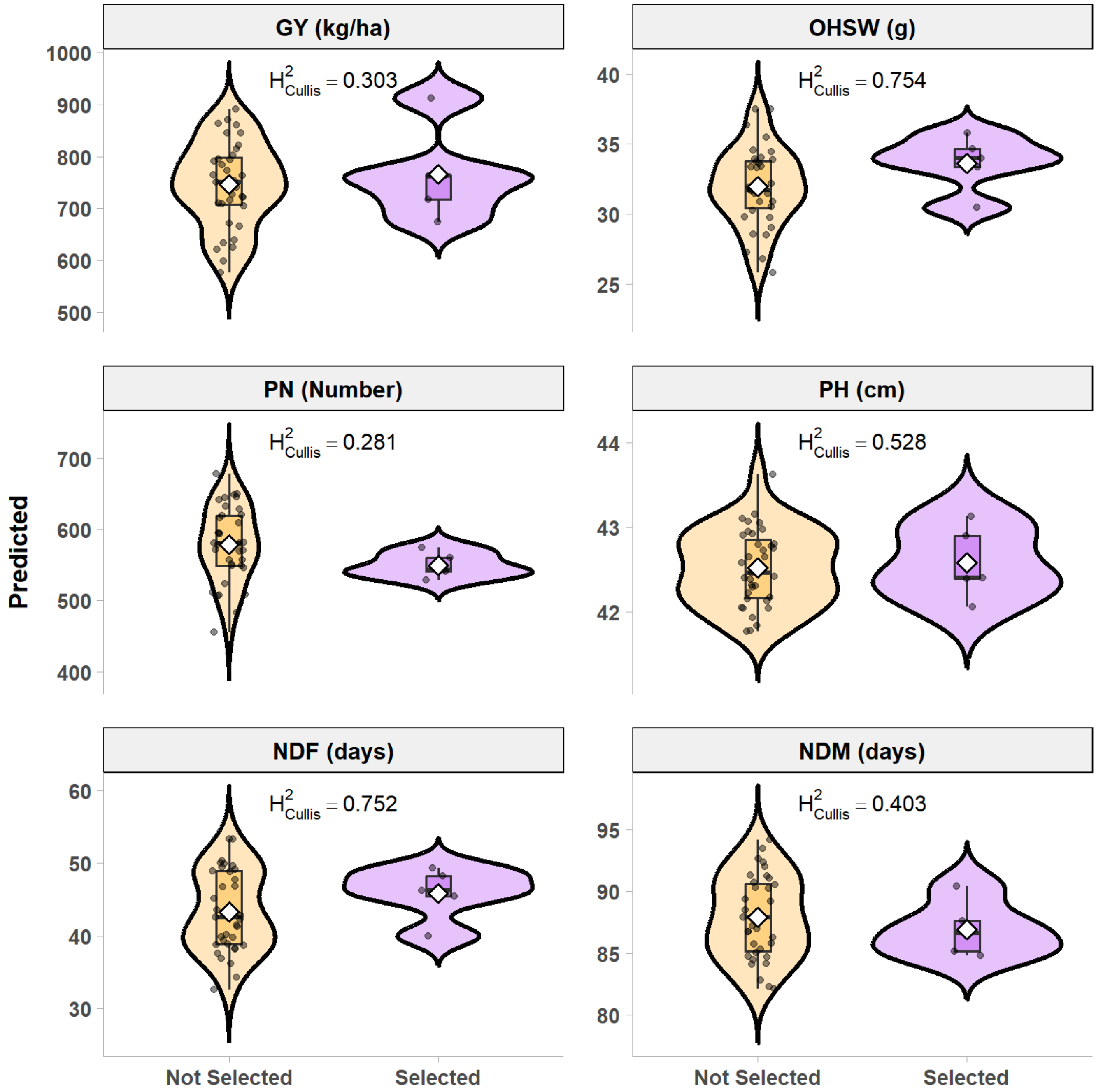
```
library(dplyr)
library(tidyr)
library(ggplot2)
library(ggridges)
trait_names <- c(
  GY_BLUP    = "GY (kg/ha)",
  OHSW_BLUP = "OHSW (g)",
  PN_BLUP   = "PN (Number)",
  PH_BLUP   = "PH (cm)",
  NDF_BLUP    = "NDF (days)",
  NDM_BLUP    = "NDM (days)")
```

```
df_long <- bind_rows(
  lapply(names(BLUPgen), function(tr) {

    BLUPgen[[tr]] %>%
      dplyr::mutate(
        Trait  = trait_names[[tr]],
        Status = ifelse(
          GEN %in% selected$GEN,
          "Selected",
          "Not Selected")) %>%
      dplyr::select(
        GEN,
        Trait,
        Status,
        Predicted)
  }))
```

```
library(ggplot2)
library(dplyr)
library()
df_long$Trait <- factor(
  df_long$Trait,
  levels = c(
    "GY (kg/ha)",
    "OHSW (g)",
    "PN (Number)",
    "PH (cm)",
    "NDF (days)",
    "NDM (days)"
  )
)
# Cullis heritability table
h2_df <- data.frame(
  Trait = c(
    "GY (kg/ha)",
    "OHSW (g)",
    "PN (Number)",
    "PH (cm)",
    "NDF (days)",
    "NDM (days)"
  ),
  H2 = c(
    0.303,
    0.754,
    0.281,
    0.528,
    0.752,
    0.403))

h2_df$Trait <- factor(
  h2_df$Trait,
  levels = levels(df_long$Trait))
# Plot
```

```
SelectionShiftPlot <- ggplot(
  df_long,
  aes(
    x = Status,
    y = Predicted,
    fill = Status)) +
  # Violin
  geom_violin(
    alpha = 0.25,
    color = "black",
    linewidth = 1.2,
    trim = FALSE) +
  # Boxplot
  geom_boxplot(
    width = 0.12,
    alpha = 0.3,
    linewidth = 0.6,
    outlier.shape = NA) +
  # Individual genotypes
  geom_jitter(
    width = 0.08,
    size = 1.5,
    alpha = 0.45,
    color = "black") +
  # Mean
  stat_summary(
    fun = mean,
    geom = "point",
    shape = 23,
    size = 3,
    fill = "white",
    color = "black",
    stroke = 1) +
  facet_wrap(
    ~Trait,
    scales = "free_y",
    ncol = 2, nrow = 3) +
  # Heritability annotation
  geom_text(
    data = h2_df,
    aes(
      x = 1.5,
      y = Inf,
      label = paste0(
        "H[Cullis]^2==",
        sprintf("%.3f", H2))),
    inherit.aes = FALSE,
    parse = TRUE,
    fontface = "bold",
    size = 4,
    vjust = 1.5) +
  scale_fill_manual(
    values = c(
      "Not Selected" = "orange",
```

```
      "Selected" = "purple")) +
  labs(
    x = NULL,
    y = "Predicted",
    fill = NULL) +
  ggdist::theme_ggdist(base_size = 14) +
  theme(
    panel.grid = element_blank(),
    strip.background = element_rect(
      fill = "grey95",
      colour = "black",
      linewidth = 0.5),
    strip.text = element_text(
      face = "bold",
      size = 12),
    axis.title.y = element_text(
      face = "bold",
      size = 13),
    axis.text.x = element_text(
      face = "bold",
      size = 11),
    axis.text.y = element_text(
      face = "bold",
      size = 11),
    legend.position = "none",
    legend.text = element_text(
      face = "bold",
      size = 11),
    panel.spacing = unit(1.2, "lines"))
SelectionShiftPlot
```




```
ggsave("SelectionShiftPlot.pdf", plot = SelectionShiftPlot,
       width = 8,
       height = 8,
       dpi = 600)
```

Density plot

```
ggsave(plot = densplot, "densplot.pdf", width = 14, height = 8, dpi = 600)
```

Environmental Similarity

```
library(EnvRtype)
library(terra)
```

```
library(raster)
library(dplyr)
library(ggplot2)
library(reshape2)

# ENVIRONMENTS
env.i <- c(
  "Piracicaba (SP)",
  "Teresina (PI)",
  "Tianguá (CE)"
)

lat <- c(-22.708333, -5.093889, -3.732222)
lon <- c(-47.636667, -42.784722, -41.012222)

plant.date <- c("2024-12-19", "2023-03-14", "2023-02-14")
harv.date  <- c("2025-04-15", "2023-07-14", "2023-06-14")

# WEATHER DATA
df.clim <- get_weather(
  env.id = env.i,
  lat = lat,
  lon = lon,
  start.day = plant.date,
  end.day = harv.date
)

# CLIMATE COVARIATES
var.clim <- c("T2M", "VPD", "PRECTOT", "GWETTOP")

df.clim$DATE <- as.Date(as.character(df.clim$YYYYMMDD), format = "%Y%m%d")
df.clim$YEAR <- as.numeric(format(df.clim$DATE, "%Y"))
df.clim$MM   <- as.numeric(format(df.clim$DATE, "%m"))
df.clim$DD   <- as.numeric(format(df.clim$DATE, "%d"))

EC <- W_matrix(
  env.data = df.clim,
  env.id = "env",
  var.id = var.clim,
  statistic = "mean"
)

EC <- data.frame(env = rownames(EC), EC, row.names = NULL)

# SOIL RASTER
soilraster <- raster(
  "C:/Users/joaop/Documents/GitHub/Kenya Maps/KenyaMaps/Soil_Raster/hwsd_domi.tif"
)

coords <- data.frame(env = env.i, lon = lon, lat = lat)

soil_values <- extract(soilraster, coords[, c("lon", "lat")])

soil_class <- data.frame(env = env.i, soil_class = soil_values)
```

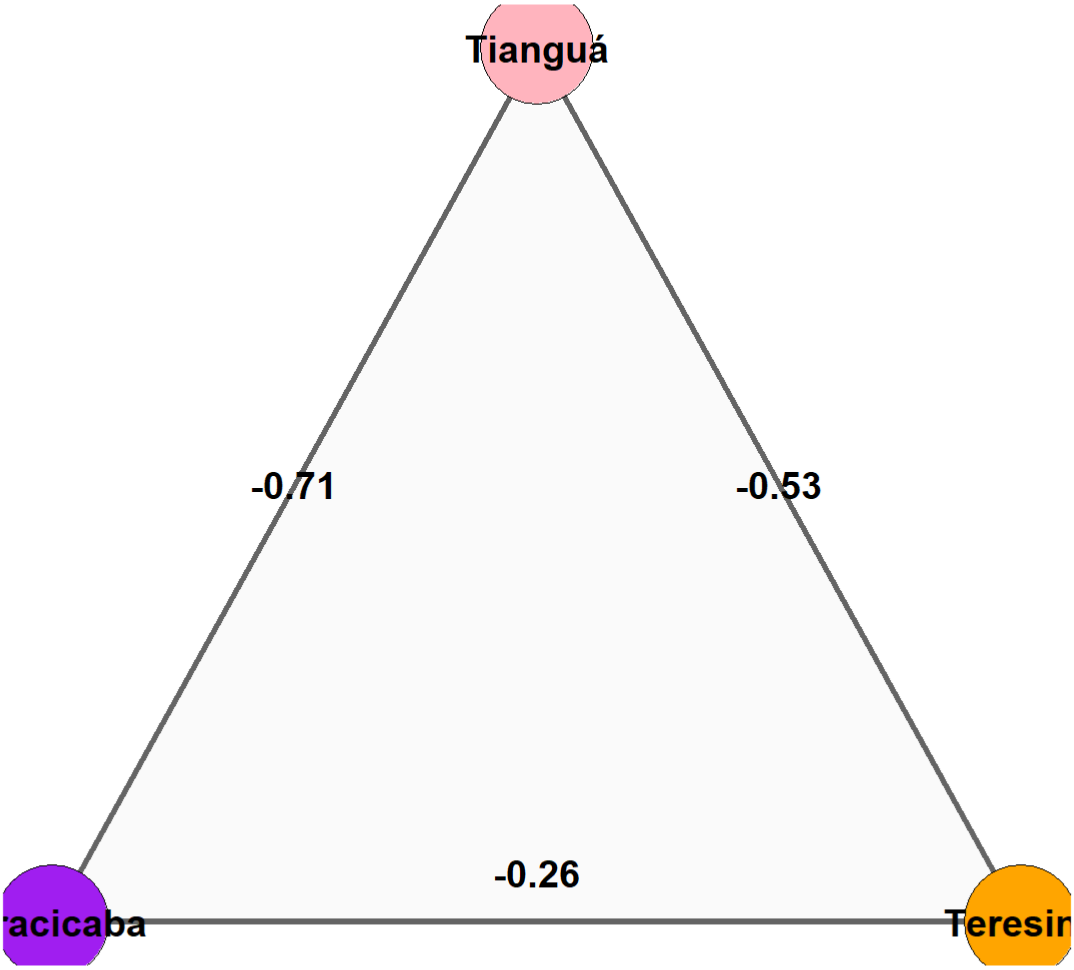
```
soil_class_names <- data.frame(
  soil_class = c(1, 10),
  class_name = c("Lixisol", "Regosol")
)

soil_class <- left_join(soil_class, soil_class_names, by = "soil_class")

# MERGE CLIMATE + SOIL
EC_soil <- left_join(
  EC,
  soil_class[, c("env", "class_name")],
  by = "env"
)

# DUMMY VARIABLES (FIXED)
soil_dummy <- model.matrix(~ class_name - 1, data = EC_soil)
soil_dummy <- as.data.frame(soil_dummy)

# FINAL MATRIX
EC_final <- cbind(
  EC_soil %>% dplyr::select(-class_name),
  soil_dummy
)

EC_numeric <- EC_final %>% dplyr::select(where(is.numeric))
EC_numeric <- as.matrix(EC_numeric)

rownames(EC_numeric) <- EC_final$env

# IMPUTE NA
for (i in 1:ncol(EC_numeric)) {
  EC_numeric[is.na(EC_numeric[, i]), i] <- mean(EC_numeric[, i], na.rm = TRUE)
}

# ENVIRONMENTAL KERNEL
K_E <- env_kernel(env.data = EC_numeric)[[2]]

# HEATMAP DATA
K_df <- melt(K_E)

colnames(K_df) <- c("Env1", "Env2", "Similarity")

K_df <- K_df %>%
  dplyr::mutate(
    Env1 = factor(Env1, levels = rownames(K_E)),
    Env2 = factor(Env2, levels = colnames(K_E))
  ) %>%
  filter(as.numeric(Env1) >= as.numeric(Env2))

# PLOT
envsim <- ggplot(K_df, aes(Env1, Env2, fill = Similarity)) +

  geom_tile(color = "white", linewidth = 0.5) +
```

```
  scale_fill_gradientn(
    colors = c("purple", "#b2abd2", "#f7f7f7", "#fdb863", "orange"),
    name = "Environmental Similarity"
  ) +

  coord_equal() +

  theme_minimal(base_size = 16) +

  theme(
    panel.grid = element_blank(),

    axis.text.x = element_text(face = "bold", size = 12),
    axis.text.y = element_text(face = "bold", size = 12),

    axis.title = element_blank(),

    plot.title = element_text(
      face = "bold",
      hjust = 0.5,
      size = 18
    ),

    legend.position = "bottom",

    legend.title = element_text(face = "bold", size = 12),
    legend.text = element_text(face = "bold", size = 10)
  )
```

```
library(igraph)
library(ggraph)
library(ggplot2)
env_df <- data.frame(
  env = c(
    "Piracicaba",
    "Teresina",
    "Tianguá"
  ),
  x = c(0, 1, 0.5),
  y = c(0, 0, 0.9)
)

envsim2 = ggplot(env_df, aes(x, y)) +

  # triangle
  geom_polygon(
    fill = "#F7F7F7",
    color = "grey40",
    linewidth = 1.2,
    alpha = 0.5
  ) +
```

```
  # environment points
  geom_point(
    aes(fill = env),
    shape = 21,
    size = 20,
    color = "black",
    stroke = 0
  ) +

  # environment labels
  geom_text(
    aes(label = env),
    fontface = "bold",
    size = 5,
    nudge_y = 0,
    nudge_x = 0
  ) +

  # similarity values
  annotate(
    "text",
    x = 0.5,
    y = 0.05,
    label = round(K_E[1,2], 2),
    size = 5,
    fontface = "bold"
  ) +

  annotate(
    "text",
    x = 0.25,
    y = 0.45,
    label = round(K_E[1,3], 2),
    size = 5,
    fontface = "bold"
  ) +

  annotate(
    "text",
    x = 0.75,
    y = 0.45,
    label = round(K_E[2,3], 2),
    size = 5,
    fontface = "bold") +

  scale_fill_manual(
    values = c(
      "Piracicaba" = "purple",
      "Teresina"   = "orange",
      "Tianguá"    = "lightpink")) +

  coord_equal() +
  theme_void() +
  theme(
```

```
    legend.position = "none"
  )
envsim2
```




```
ggsave("envsim2.pdf", plot = envsim2,
       width = 8,
       height = 8,
       dpi = 600)
```